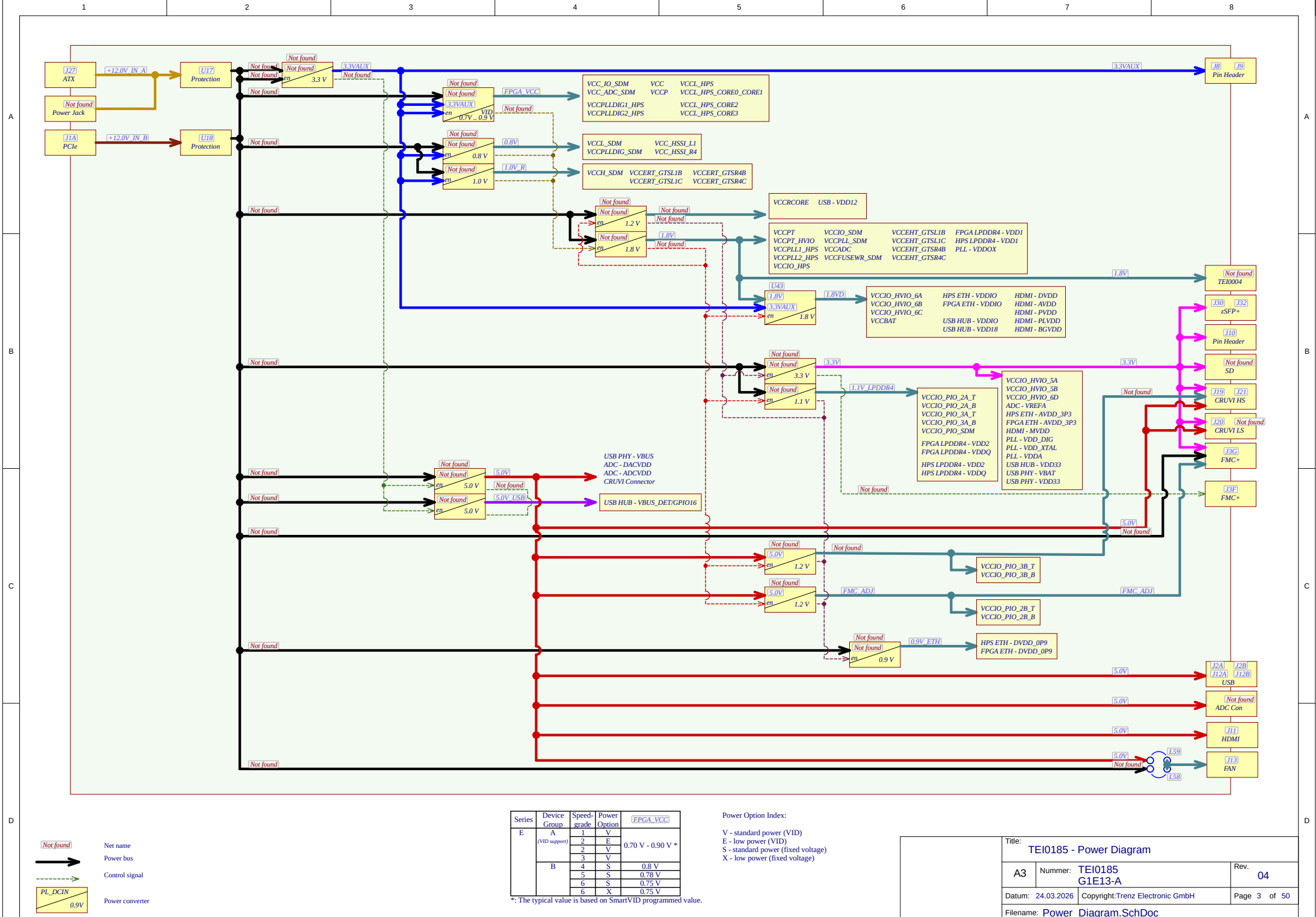


1		2		3		4	
A	REV		Description				
	-01		Initial revision			VT	
	-02		1. Change J29 power connector 2. Added clock signal 25MHz HPS_OSC_CLK1 to HPS_IOA_12 3. Added series resistor R394 to HPS_OSC_CLK1 4. Changed resistors value R169, R172, R179, R190 33R to 100R 5. DIPSW1 connected from HPS_IO12 to HPS_IO2 6. Added additional clock signal FPGA_25M_CLK to FPGA (pin U1.A23). Net Not found (pin U1.A23) connect to pin U1.B14 7. Added series resistor R396 to FPGA_25M_CLK 8. Added resistor Not found (set as DNP) 9. Added ability to recognise PCB version (pin U1.A8 is connected to GND)			VT	
B	-03		1. Swapped nets ETH1_RXCK and ETH1_RXD1 2. Changed voltage 1.2V to VIO_CRUVI on the pull-up R371, R374 3. Fixed I2C connection on the I2C Mux (U57): SFPA_SDA<->SFPB_SCL, SFPB_SDA<->SFPB_SCL 4.Added additional resistors R397, R398 (assembly option) for select voltage VCCIO_PIO_SDM 5.Added additional resistors R399,R400,R401 to unsued JTAG FMC+ 6. Changed capacitor size 0201 to 0603 for C159 7. Changed PCB revision recognition (pin U1.A8 is set as NC, pin U1.B4 is connected to GND)			VT	
	-04		1. Swapped RGMII nets in bank 6C as recommended Altera specification (ug-813266, p17)			VT	
C							
D							
1		2		3		4	

Title: TE0185 - Revision Changes		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 2 of 50
Filename: Revision Changes.SchDoc		

Title: TEI0185 - Revision Changes		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 2 of 50
Filename: Revision Changes.SchDoc		



Series	Device Group	Speed-grade	Power Option	<i>(EPGA VCC)</i>
E	A (VFD support)	1	V	0.70 V - 0.90 V
		2	E	
		2	V	
		3	V	
	B	4	S	0.8 V
		5	S	0.78 V
		6	S	0.75 V
		6	X	0.75 V

*: The typical value is based on SmartVID programmed value.

Power Option Index:

- V - standard power (VID)
- E - low power (VID)
- S - standard power (fixed voltage)
- X - low power (fixed voltage)

Title:	
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TEI0185 - Power Diagram

A3

Nummer:	TEI0185 G1E13-A
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Rev. 04

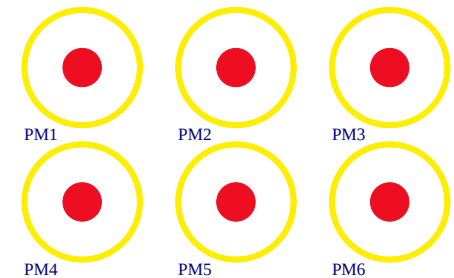
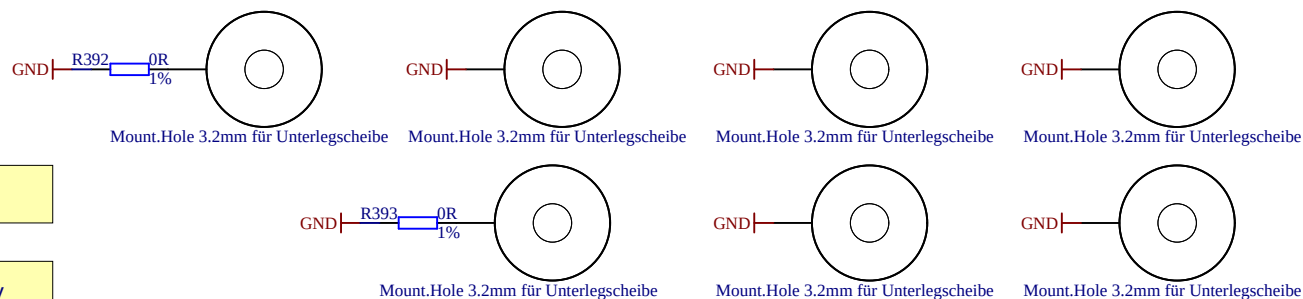
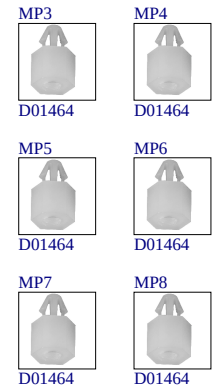
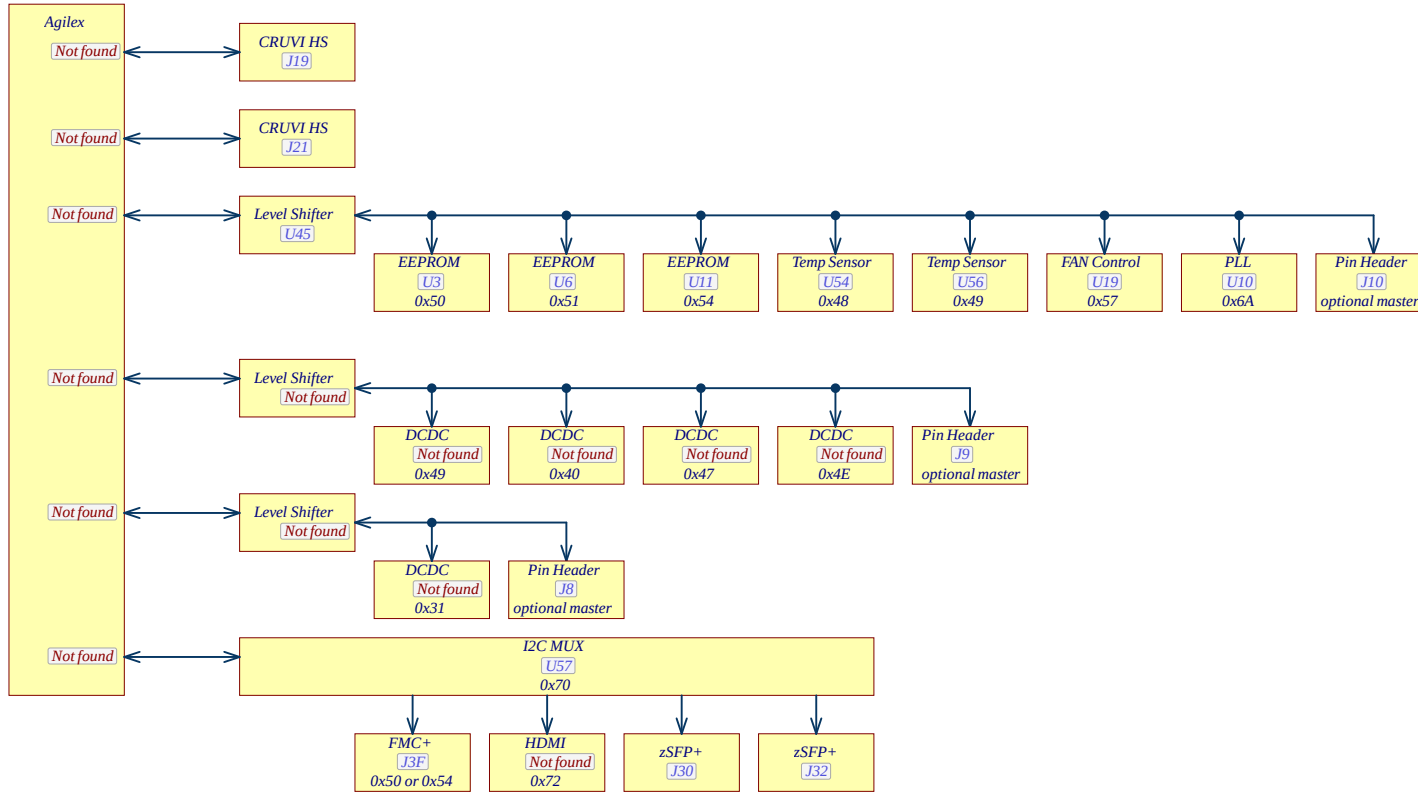
Datum: 24.03.2026

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Filename: Power_Diagram.SchDoc

I2C Structure:



CE

CE Logo on Top Overlay

CE-TOPOVERLAY
RoHS

RoHS Logo on Top Overlay

RoHS-TOPOVERLAY
WEEE

WEEE Logo on Top Overlay

WEEE-TOPOVERLAY
Serial1
Serial
Serialnumber 6,3 x 6.3mm

Design drawn by: VT

Checked by:

Assembly variant: [G1E13-A](#)

Created by:

Modified by:

Modified at:

Title:	TEI0185
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A4

Number:	TEI0185 G1E13-A
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Rev.	04
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Date: 24.03.2026

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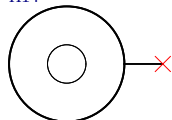
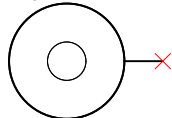
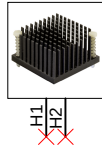
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	1	2	3	4
A	H14  Mount.Hole 3.2mm (unplated)			
B	H16  Mount.Hole 3.2mm (unplated)	KK1  Heatsink HSB43-454515P		
C				
D			Title: FPGA A4 Number: TEI0185 G1E13-A Rev. 04 Date: 24.04.2026 Copyright: Trenz Electronic GmbH Page 5 of 50 Filename: FPGA.SchDoc	
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A

A

B

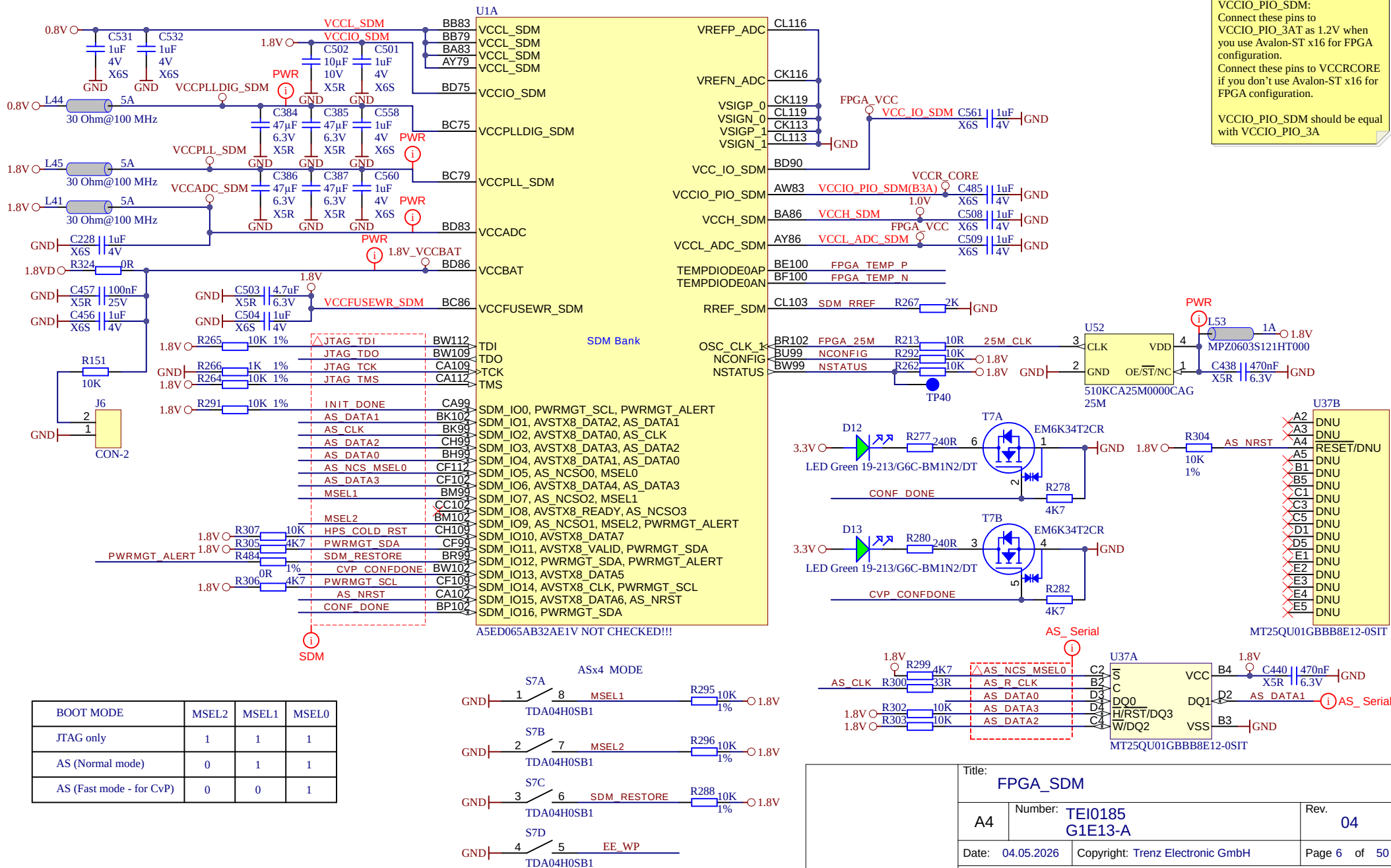
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C

D

D

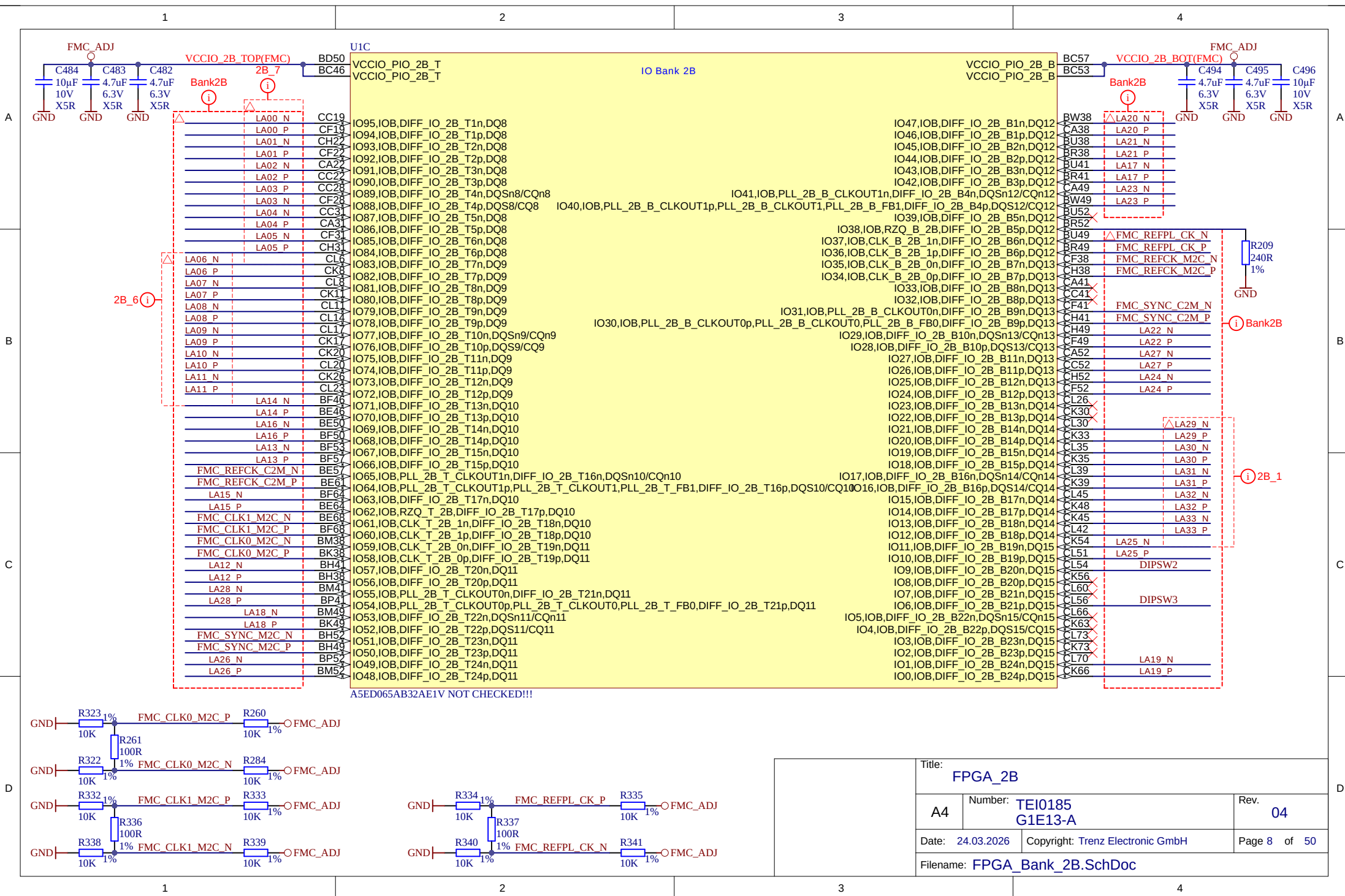


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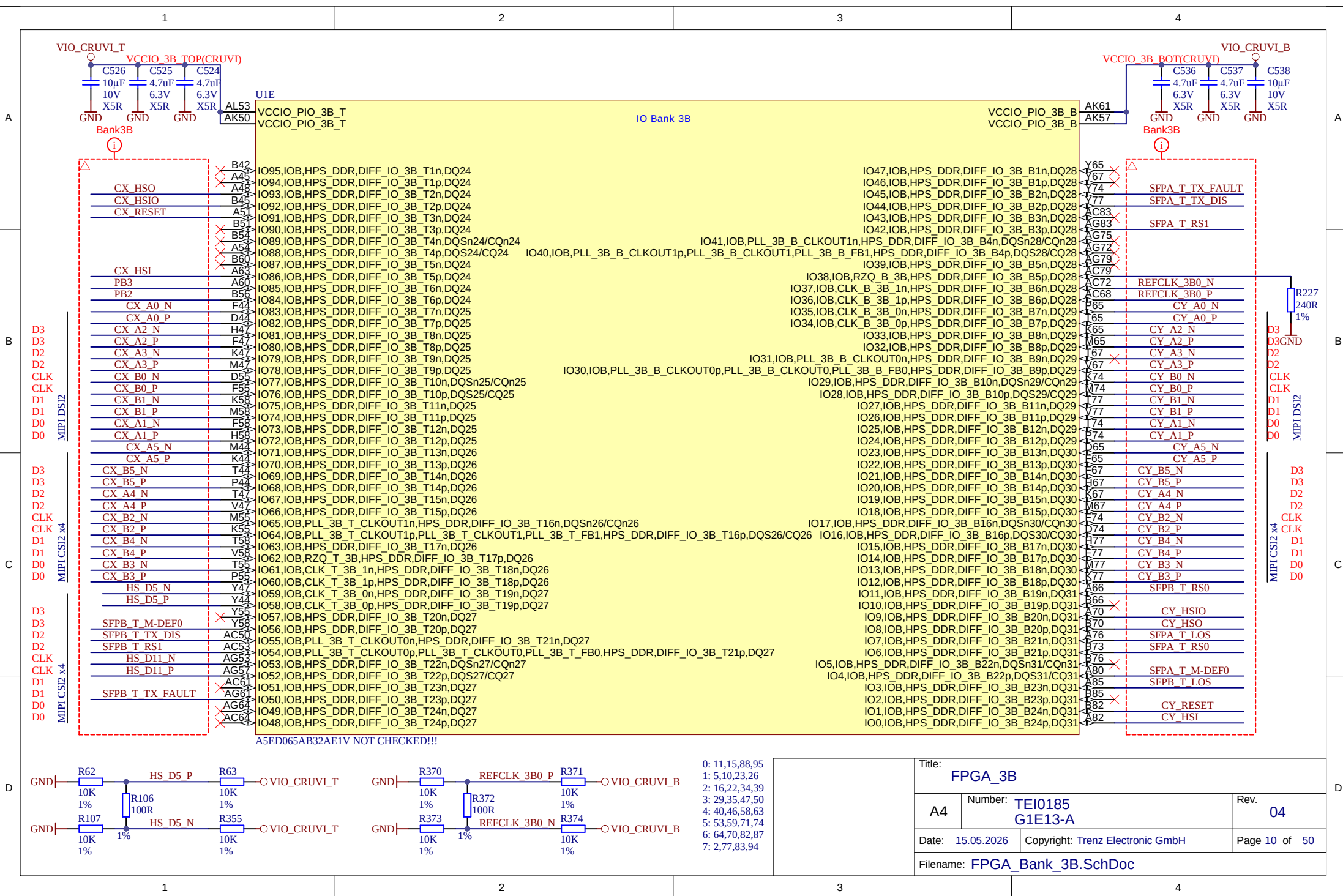
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1

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1.8VIO

HPS
i

U1G

HPS_OSC_CLK
DIPSW1
UART0_TX
UART0_RX
I2C0_SDA
GPIO_LED0
GPIO_LED1
PB0
PB1
DIPSW0
HPS_OSC_CLK1
USB_CLK
USB_STP
USB_DIR
USB_DATA0
USB_DATA1
USB_NXT
USB_DATA2
USB_DATA3
USB_DATA4
USB_DATA5
USB_DATA6
USB_DATA7

W135 HPS_IOA_1,GPIO0_IO0,SPIM0_SS1_N,SPIS0_CLK,UART0_CTS_N,NAND_ADQ0,SDMMC_DATA0,USB0_CLK,EMAC0_PPS0,TRACE_D10
U135 HPS_IOA_2,GPIO0_IO1,SPIM1_SS1_N,SPIS0_MOSI,UART0_RTS_N,NAND_ADQ1,SDMMC_DATA1,USB0_STP,EMAC0_PPSTRIG0,TRACE_D9
W134 HPS_IOA_3,GPIO0_IO2,SPIS0_SS0_N,UART0_TX,I2C1_SDA,NAND_WE_N,SDMMC_CLK,USB0_DIR,EMAC1_PPS1,TRACE_D8
U134 HPS_IOA_4,GPIO0_IO3,SPIS0_MISO,UART0_RX,I2C1_SCL,NAND_RE_N,USB0_DATA0,EMAC1_PPSTRIG1,TRACE_D7
AL120 HPS_IOA_5,GPIO0_IO4,SPIM0_CLK,UART1_CTS_N,I2C0_SDA,NAND_WP_N,SDMMC_WRITE_PROTECT,USB0_DATA1,EMAC2_PPS2,TRACE_D6
R134 HPS_IOA_6,GPIO0_IO5,SPIM0_MOSI,UART1_RTS_N,I2C0_SCL,NAND_ADQ2,SDMMC_DATA2,USB0_NXT,EMAC2_PPSTRIG2,TRACE_D5
AG115 HPS_IOA_7,GPIO0_IO6,SPIM0_MISO,MDIO2_MDIO,UART1_TX,I2C2_EMAC2_SDA,NAND_ADQ3,SDMMC_DATA3,USB0_DATA2,TRACE_D4
N135 HPS_IOA_8,GPIO0_IO7,SPIM0_SS0_N,MDIO2_MDC,UART1_RX,I2C2_EMAC2_SCL,NAND_CLE,SDMMC_CMD,USB0_DATA3,TRACE_D15
AK120 HPS_IOA_9,GPIO0_IO8,SPIM1_CLK,SPIS1_CLK,MDIO1_MDIO,I2C2_EMAC1_SDA,NAND_ADQ4,SDMMC_DATA4,USB0_DATA4,I3C1_SDA,TRACE_D14
N135 HPS_IOA_10,GPIO0_IO9,SPIM1_MOSI,SPIS1_MOSI,MDIO1_MDC,I2C2_EMAC1_SCL,NAND_ADQ5,SDMMC_DATA5,USB0_DATA5,I3C1_SCL,TRACE_D13
T135 HPS_IOA_11,GPIO0_IO10,SPIM1_MISO,SPIS1_SS0_N,MDIO0_MDIO,I2C2_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,USB0_DATA6,I3C0_SDA,TRACE_D12
L135 HPS_IOA_12,GPIO0_IO11,SPIM1_SS0_N,SPIS1_MISO,MDIO0_MDC,I2C2_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,USB0_DATA7,I3C0_SCL,TRACE_D11
P135 HPS_IOA_13,GPIO0_IO12,NAND_ALE,SDMMC_PU_PD_DATA2,USB1_CLK,EMAC0_TX_CLK,TRACE_D10
L135 HPS_IOA_14,GPIO0_IO13,NAND_RB_N,SDMMC_PWR_ENA,USB1_STP,EMAC0_RX_CTL,TRACE_D9
J135 HPS_IOA_15,GPIO0_IO14,NAND_CE_N,USB1_DIR,EMAC0_RX_CLK,TRACE_D8
AD135 HPS_IOA_16,GPIO0_IO15,NAND_DQS,SDMMC_DATA_STROBE,USB1_DATA0,EMAC0_RX_CTL,TRACE_D7
M135 HPS_IOA_17,GPIO0_IO16,I3C1_SDA,NAND_ADQ8,USB1_DATA1,EMAC0_TXD0,TRACE_D6
K135 HPS_IOA_18,GPIO0_IO17,I3C1_SCL,NAND_ADQ9,USB1_NXT,EMAC0_TXD1,TRACE_D5
AG120 HPS_IOA_19,GPIO0_IO18,I3C0_SDA,NAND_ADQ10,USB1_DATA2,EMAC0_RXD0,TRACE_D4
J135 HPS_IOA_20,GPIO0_IO19,SPIM1_SS1_N,I3C0_SCL,NAND_ADQ11,USB1_DATA3,EMAC0_RXD1,TRACE_CLK
AG120 HPS_IOA_21,GPIO0_IO20,SPIM1_CLK,SPIS0_CLK,UART0_CTS_N,I2C1_SDA,NAND_ADQ12,USB1_DATA4,EMAC0_TXD2,TRACE_D0
G135 HPS_IOA_22,GPIO0_IO21,SPIM1_MOSI,SPIS0_MOSI,UART0_RTS_N,I2C1_SCL,NAND_ADQ13,USB1_DATA5,EMAC0_TXD3,TRACE_D1
G135 HPS_IOA_23,GPIO0_IO22,SPIM1_MISO,SPIS0_SS0_N,UART0_TX,I2C0_SDA,NAND_ADQ14,USB1_DATA6,EMAC0_RXD2,TRACE_D2
G135 HPS_IOA_24,GPIO0_IO23,SPIM1_SS0_N,SPIS0_MISO,UART0_RX,I2C0_SCL,NAND_ADQ15,USB1_DATA7,EMAC0_RXD3,TRACE_D3

HPS Bank IOA

A5ED065AB32AE1V NOT CHECKED!!!

U1H

SD_RDAT0 R8 33R 1% SD_DAT0
SD_RDAT1 R360 33R 1% SD_DAT1
SD_RCLK R361 33R 1% SD_CLK
USB_RST
SD_RDAT2 R362 33R 1% SD_DAT2
SD_RDAT3 R363 33R 1% SD_DAT3
SD_RCMD R364 33R 1% SD_CMD
ETH_MDIO
ETH_MDC
ETH_RST
SD_VSEL_1V8
ETH_TXCK
ETH_TXCTL
ETH_RXCK
ETH_RXCTL
ETH_TXD0
ETH_TXD1
ETH_RXD0
ETH_RXD1
ETH_TXD2
ETH_TXD3
ETH_RXD2
ETH_RXD3

E135 HPS_IOB_1,GPIO1_IO0,SPIM1_CLK,UART0_CTS_N,EMAC0_PPS0,NAND_ADQ0,SDMMC_DATA0,I3C0_SDA_PULLUP_EN,EMAC1_TX_CLK,TRACE_D10
F135 HPS_IOB_2,GPIO1_IO1,SPIM1_MOSI,UART0_RTS_N,EMAC0_PPSTRIG0,NAND_ADQ1,SDMMC_DATA1,I3C1_SDA_PULLUP_EN,EMAC1_TX_CTL,TRACE_D9
D135 HPS_IOB_3,GPIO1_IO2,SPIM1_MISO,UART0_TX,I2C0_SDA,NAND_WE_N,SDMMC_CLK,EMAC1_RX_CLK,TRACE_D8
B135 HPS_IOB_4,GPIO1_IO3,SPIM1_SS0_N,UART0_RX,I2C0_SCL,NAND_RE_N,EMAC1_RX_CTL,TRACE_D7
AA135 HPS_IOB_5,GPIO1_IO4,SPIM1_SS1_N,SPIS1_CLK,UART1_CTS_N,EMAC2_PPS2,NAND_WP_N,SDMMC_WRITE_PROTECT,I3C1_SDA,EMAC1_TXD0,TRACE_D6
V125 HPS_IOB_6,GPIO1_IO5,SPIS1_MOSI,UART1_RTS_N,EMAC2_PPSTRIG2,NAND_ADQ2,SDMMC_DATA2,I3C1_SCL,EMAC1_TXD1,TRACE_D5
AB135 HPS_IOB_7,GPIO1_IO6,SPIS1_SS0_N,UART1_TX,I2C1_SDA,NAND_ADQ3,SDMMC_DATA3,I3C0_SDA,EMAC1_RXD0,TRACE_D4
T125 HPS_IOB_8,GPIO1_IO7,SPIS1_MISO,UART1_RX,I2C1_SCL,NAND_CLE,SDMMC_CMD,I3C0_SCL,EMAC1_RXD1,TRACE_D15
Y135 HPS_IOB_9,GPIO1_IO8,JTAG_TCK,SPIS0_CLK,MDIO2_MDIO,I2C2_EMAC2_SDA,NAND_ADQ4,SDMMC_DATA4,EMAC1_TXD2,TRACE_D14
T125 HPS_IOB_10,GPIO1_IO9,JTAG_TMS,SPIS0_MOSI,MDIO2_MDC,I2C2_EMAC2_SCL,NAND_ADQ5,SDMMC_DATA5,EMAC1_TXD3,TRACE_D13
P125 HPS_IOB_11,GPIO1_IO10,JTAG_TDO,SPIS0_SS0_N,MDIO0_MDIO,I2C2_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,EMAC1_RXD2,TRACE_D12
K125 HPS_IOB_12,GPIO1_IO11,JTAG_TDI,SPIS0_MISO,MDIO0_MDC,I2C2_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,EMAC1_RXD3,TRACE_D11
M125 HPS_IOB_13,GPIO1_IO12,I2C1_SDA,NAND_ALE,SDMMC_PU_PD_DATA2,EMAC2_TX_CLK,TRACE_D10
K125 HPS_IOB_14,GPIO1_IO13,I2C1_SCL,NAND_RB_N,SDMMC_PWR_ENA,EMAC2_TX_CTL,TRACE_D9
AB125 HPS_IOB_15,GPIO1_IO14,UART1_TX,NAND_CE_N,I3C1_SDA,EMAC2_RX_CLK,TRACE_D8
K125 HPS_IOB_16,GPIO1_IO15,UART1_RX,NAND_DQS,SDMMC_DATA_STROBE,I3C1_SCL,EMAC2_RX_CTL,TRACE_D7
Y125 HPS_IOB_17,GPIO1_IO16,UART1_CTS_N,NAND_ADQ8,I3C0_SDA,EMAC2_TXD0,TRACE_D6
H125 HPS_IOB_18,GPIO1_IO17,SPIM0_SS1_N,UART1_RTS_N,NAND_ADQ9,I3C0_SCL,EMAC2_TXD1,TRACE_D5
AB125 HPS_IOB_19,GPIO1_IO18,SPIM0_MISO,MDIO1_MDIO,I2C2_EMAC1_SDA,NAND_ADQ10,EMAC2_RXD0,TRACE_D4
F125 HPS_IOB_20,GPIO1_IO19,SPIM0_SS0_N,MDIO1_MDC,I2C2_EMAC1_SCL,NAND_ADQ11,EMAC2_RXD1,TRACE_CLK
Y125 HPS_IOB_21,GPIO1_IO20,SPIM0_CLK,SPIS1_CLK,I2C2_EMAC2_SDA,NAND_ADQ12,I3C0_SDA_PULLUP_EN,EMAC2_TXD2,TRACE_D0
Y125 HPS_IOB_22,GPIO1_IO21,SPIM0_MOSI,SPIS1_MOSI,I2C2_EMAC2_SCL,NAND_ADQ13,I3C1_SDA_PULLUP_EN,EMAC2_TXD3,TRACE_D1
F125 HPS_IOB_23,GPIO1_IO22,SPIM0_MISO,SPIS1_SS0_N,MDIO0_MDIO,I2C2_EMAC0_SDA,NAND_ADQ14,EMAC2_RXD2,TRACE_D2
D125 HPS_IOB_24,GPIO1_IO23,SPIM0_SS0_N,SPIS1_MISO,MDIO0_MDC,I2C2_EMAC0_SCL,NAND_ADQ15,EMAC2_RXD3,TRACE_D3

HPS Bank IOB

A5ED065AB32AE1V NOT CHECKED!!!

R394

10R

1%

R212

10R

1%

25M CK

U51

CLK

VDD

4

GND

OE/ST/NC

1

510KCA25M0000CAG

25M

PWR

i

C437

470nF

6.3V

X5R

GND

L52

MPZ0603S121HT000

1A

0.18V

Title:

FPGA_HPS

A4

Number:

TEI0185

G1E13-A

Rev.

04

Date: 24.03.2026

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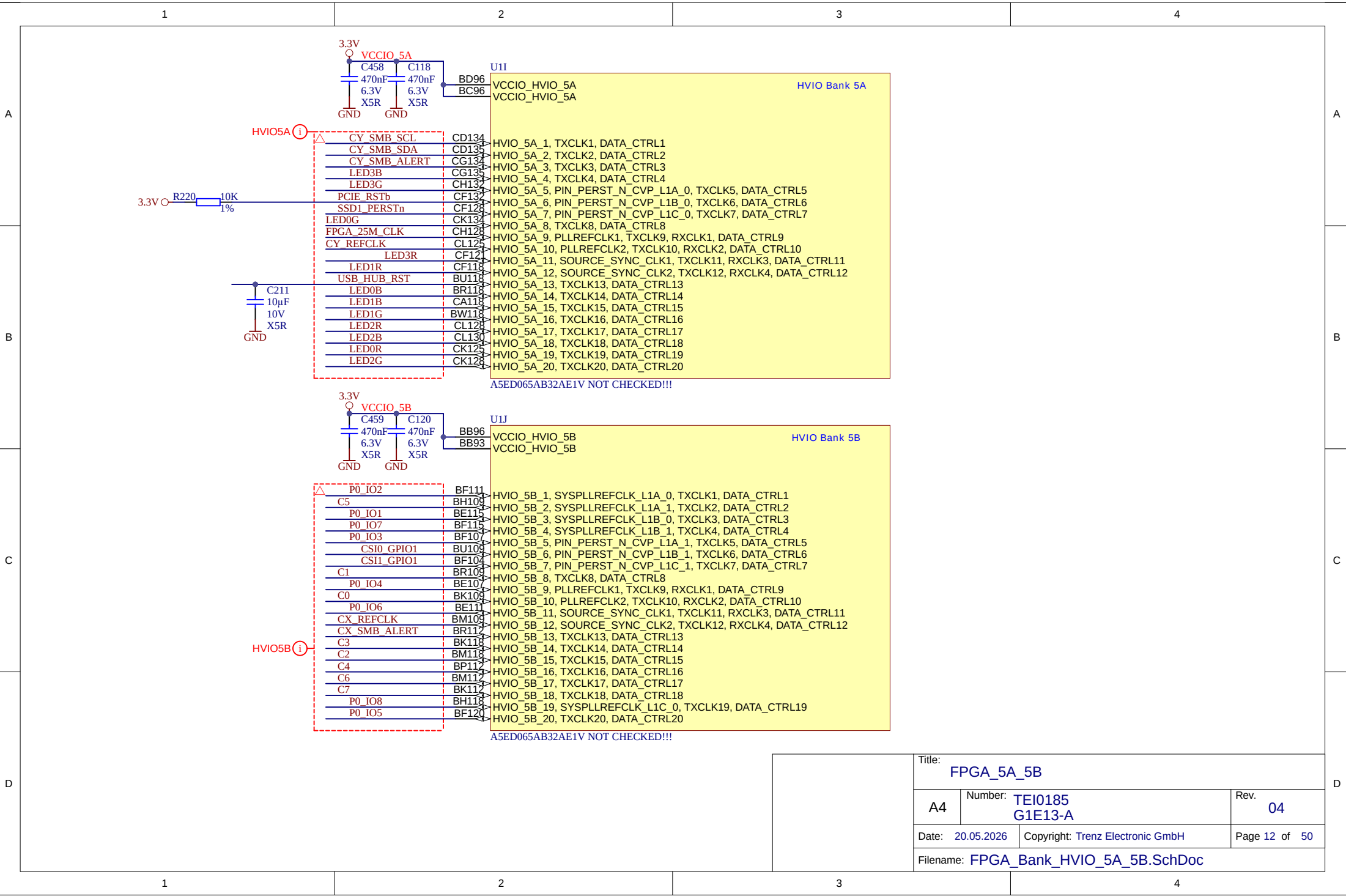
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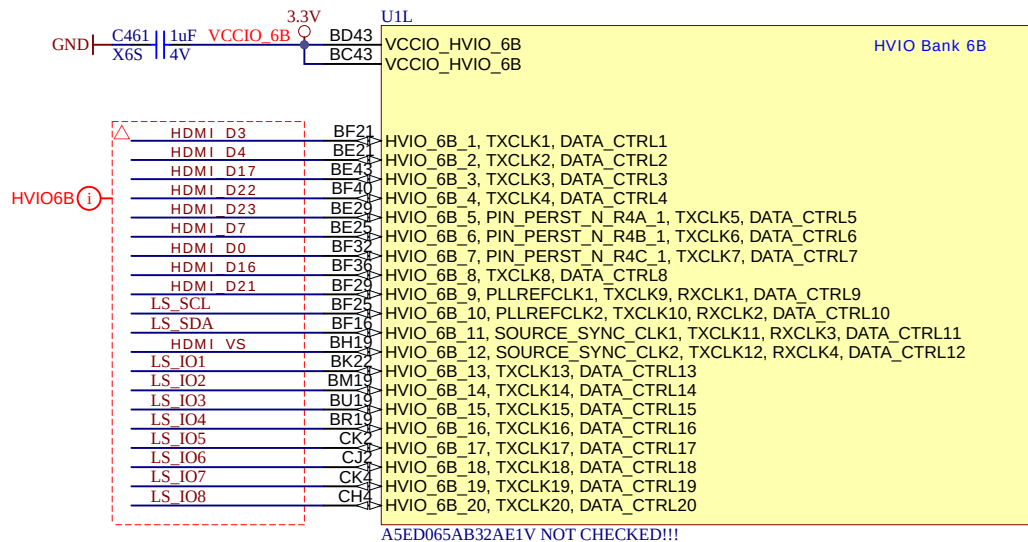
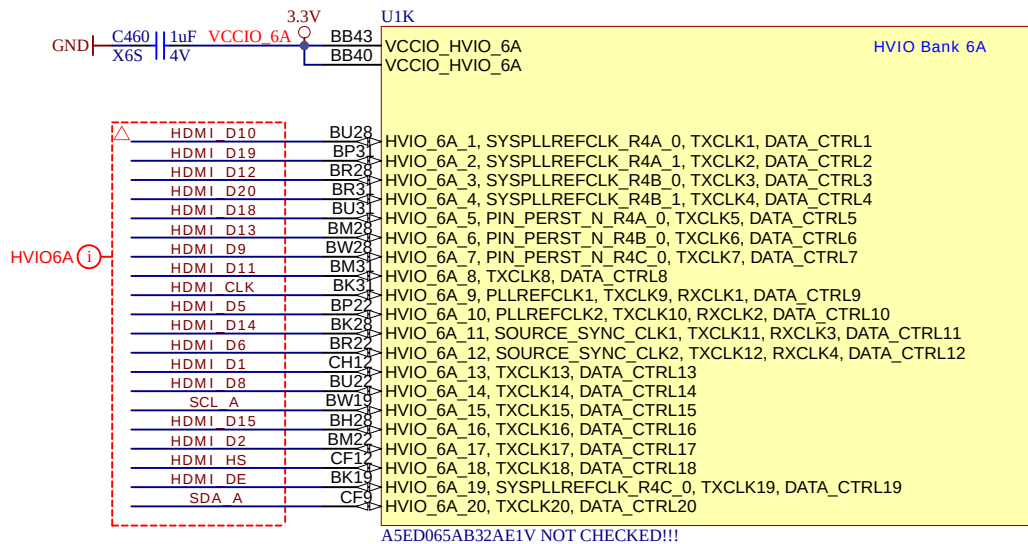
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Date: 20.05.2026	Copyright: Trenz Electronic GmbH		Page 12 of 50
Filename: FPGA_Bank_HVIO_5A_5B.SchDoc			



Title:

FPGA_6A_6B

A4

Number: TEI0185
G1E13-A

Rev.

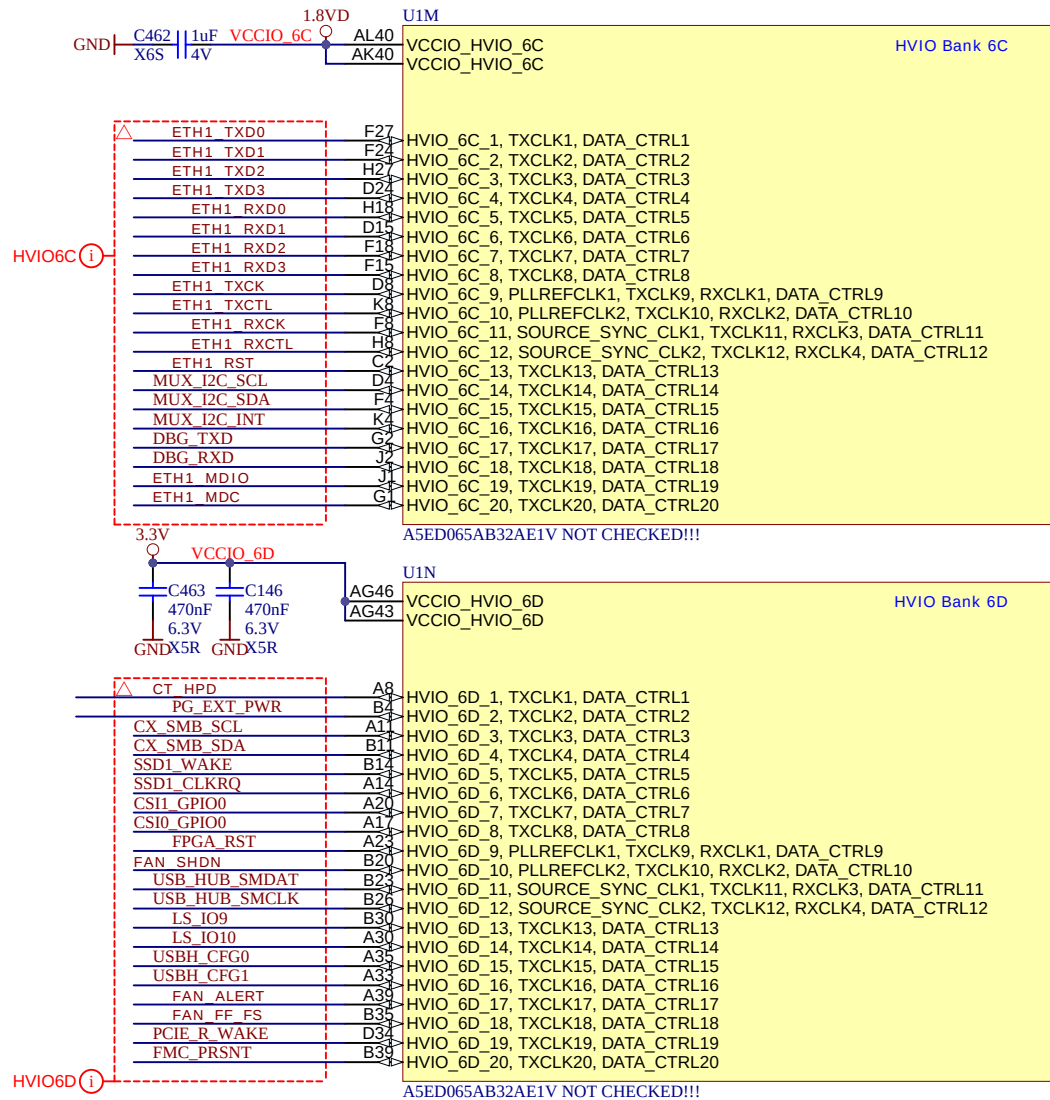
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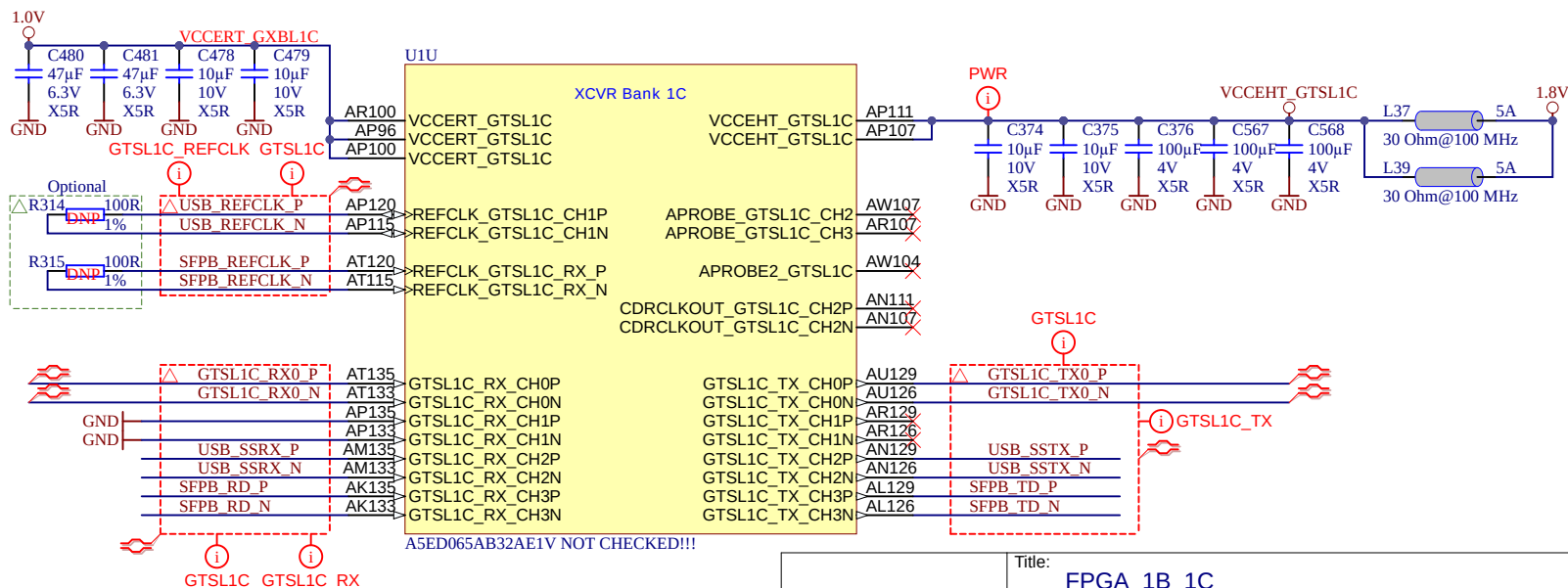
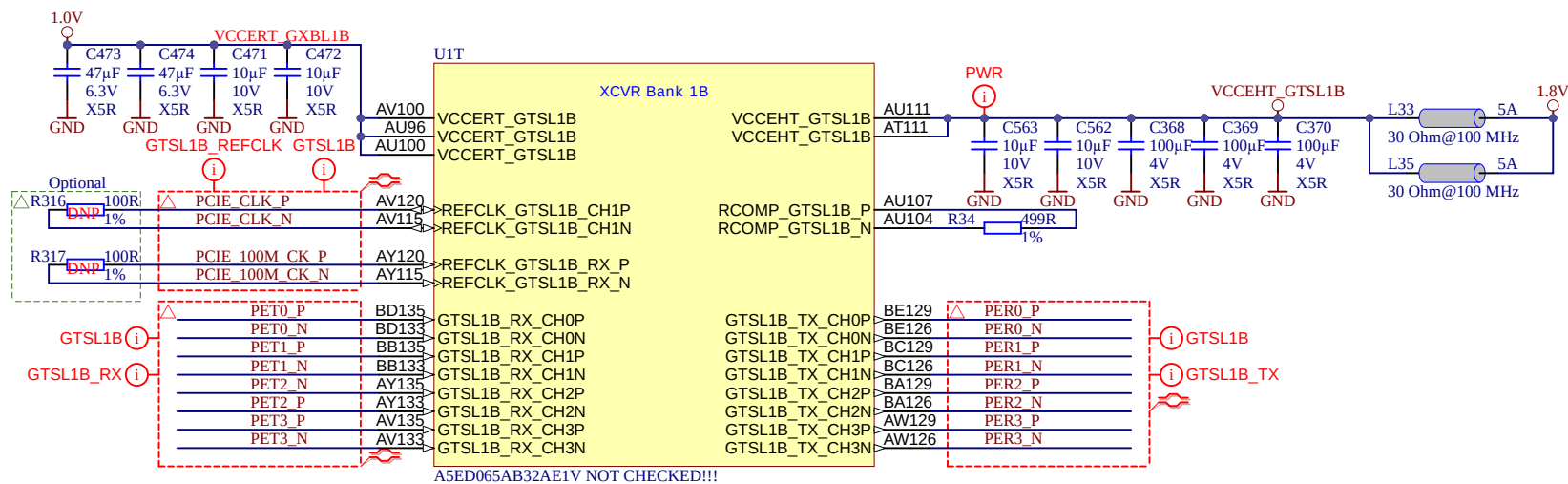
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Date: 15.05.2026	Copyright: Trenz Electronic GmbH	Page 14 of 50
Filename: FPGA_Bank_HVIO_6C_6D.SchDoc		



Title:

FPGA_1B_1C

A4

Number: TEI0185
G1E13-A

Rev.

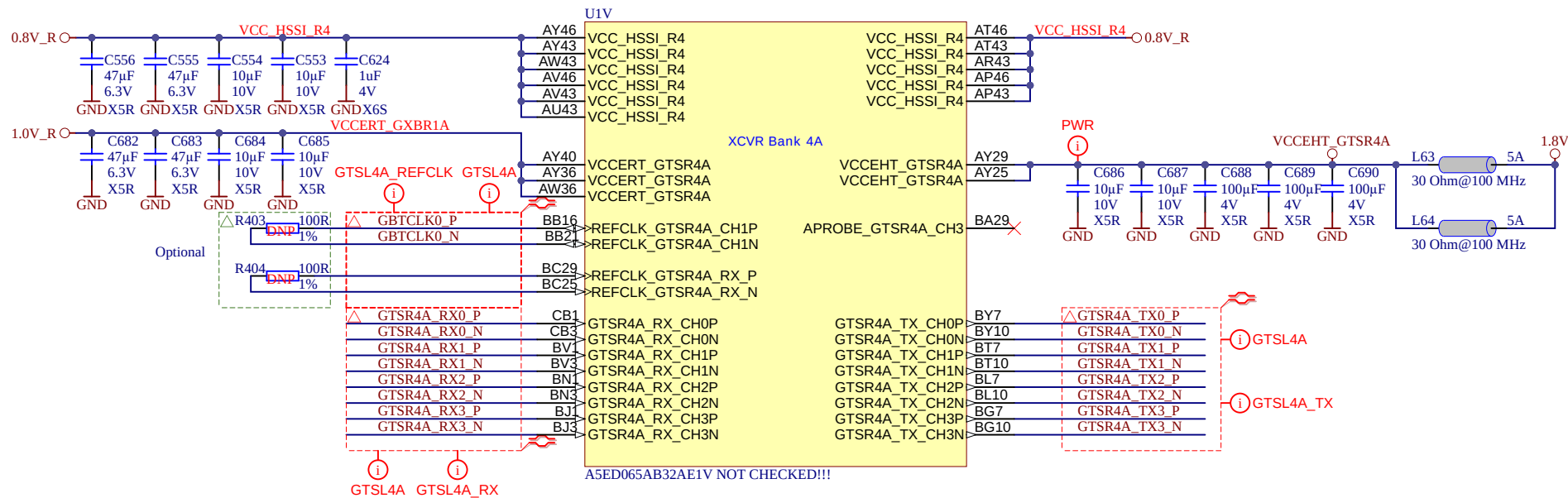
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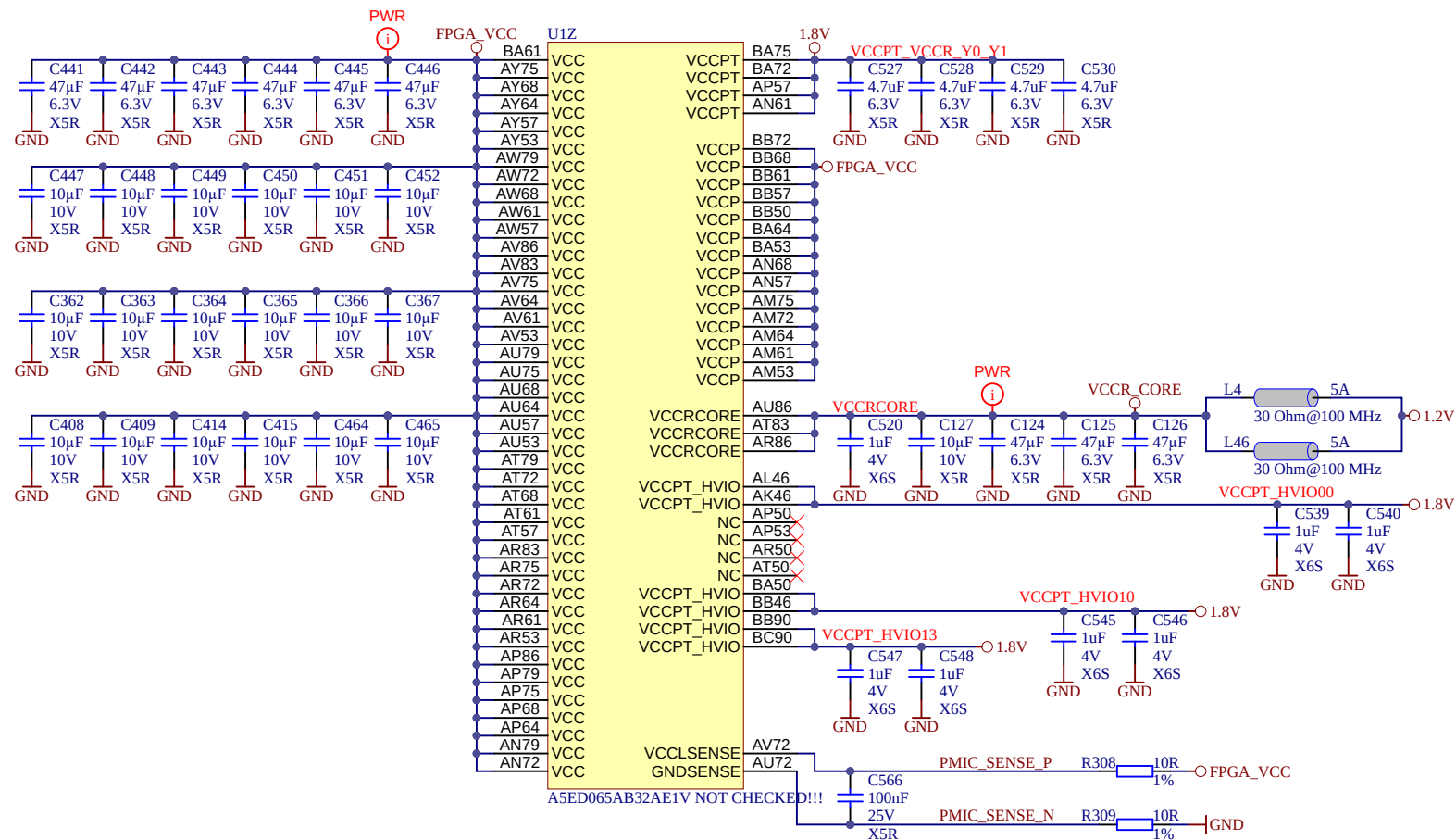
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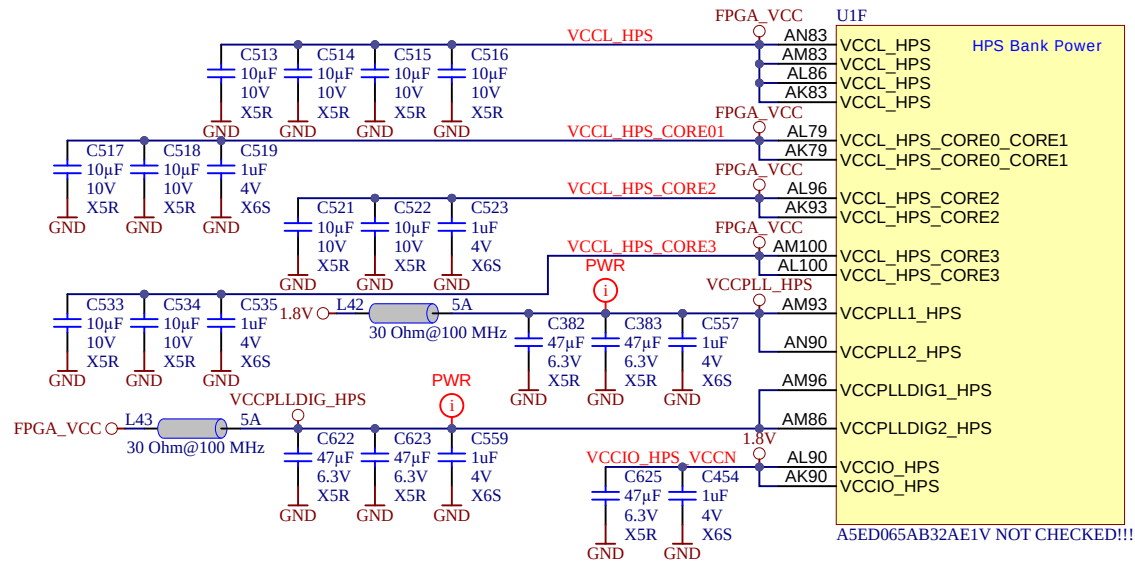
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Date: 24.04.2026	Copyright: Trenz Electronic GmbH	Page 17 of 50
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Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 19 of 50
Filename: FPGA_PWR.SchDoc		



Title: FPGA_Power		
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Date: 24.03.2026	Copyright: Trenz Electronic GmbH	
Filename: FPGA_PWR2.SchDoc		Page 20 of 50

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A

A

B

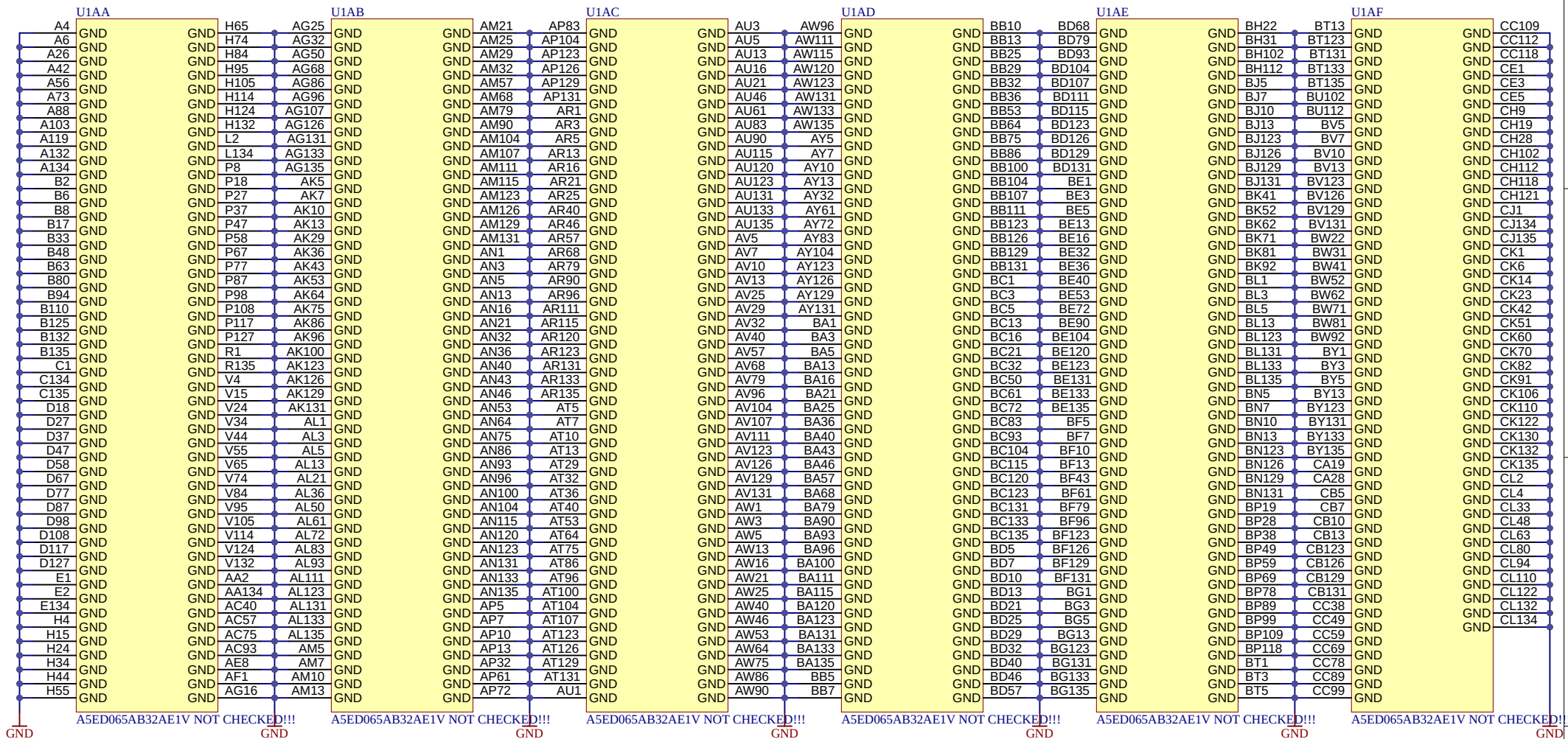
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D



Title: FPGA_GND		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 21 of 50
Filename: FPGA_GND.SchDoc		

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	1	2	3	4
A	U1O AW50 NC AY50 NC AB18 NC AB15 NC Y18 NC AG13 NC P15 NC T15 NC AH8 NC T18 NC V18 NC M18 NC Y15 NC K18 NC AH4 NC Y8 NC AB8 NC M8 NC T8 NC V8 NC M15 NC K15 NC HVIO Bank 6E A5ED065AB32AE1V NOT CHECKED!!! U1R AM43 NC AM46 NC AL16 NC AK16 NC K24 NC M24 NC M27 NC K27 NC T24 NC P24 NC V27 NC T27 NC Y24 NC AB24 NC AB27 NC Y27 NC AL25 NC AK25 NC AG21 NC AK21 NC AL29 NC AG29 NC HVIO Bank 6G A5ED065AB32AE1V NOT CHECKED!!!	U1P AU50 NC AV50 NC AD1 NC AJ1 NC AA1 NC AJ2 NC AD2 NC AF2 NC AB4 NC AE4 NC Y4 NC W2 NC W1 NC U1 NC R2 NC U2 NC N2 NC T4 NC L1 NC N1 NC M4 NC P4 NC HVIO Bank 6F A5ED065AB32AE1V NOT CHECKED!!! U1Q AM50 NC AN50 NC AK32 NC AC43 NC AG36 NC AG40 NC AL32 NC T37 NC AC46 NC T34 NC V37 NC AC36 NC Y34 NC Y37 NC M37 NC M34 NC K37 NC P34 NC K34 NC H37 NC F34 NC F37 NC HVIO Bank 6G A5ED065AB32AE1V NOT CHECKED!!!	U1Y AL43 DNU AL57 DNU AL68 DNU AL104 DNU AL107 NC AL115 DNU AM16 NC AM36 DNU AM40 DNU AM120 DNU AR32 DNU AR104 DNU BA32 DNU BA104 DNU DNU DNU DNU NC DNU NC DNU DNU DNU DNU DNU DNU DNU DNU DNU DNU NC DNU DNU DNU DNU DNU DNU DNU DNU DNU BC36 BC40 BC100 BD16 BD36 BD53 BD64 BD100 BD120 CK101 CK103 CL101 CL106 A5ED065AB32AE1V NOT CHECKED!!!	
B				
C				
D			Title: FPGA_GND A4 Number: TEI0185 G1E13-A Rev. 04 Date: 24.03.2026 Copyright: Trenz Electronic GmbH Page 22 of 50 Filename: FPGA_NC.SchDoc	
	1	2	3	4

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A

A

B

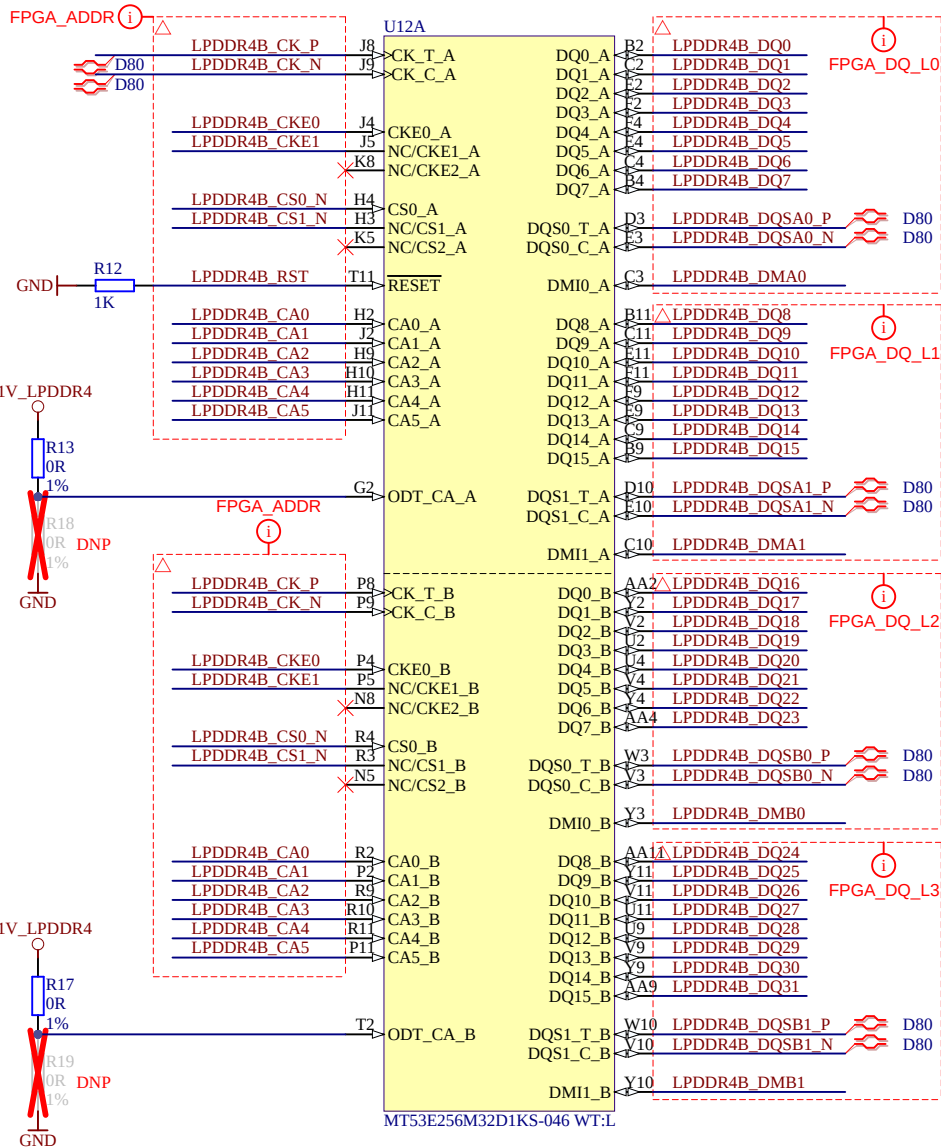
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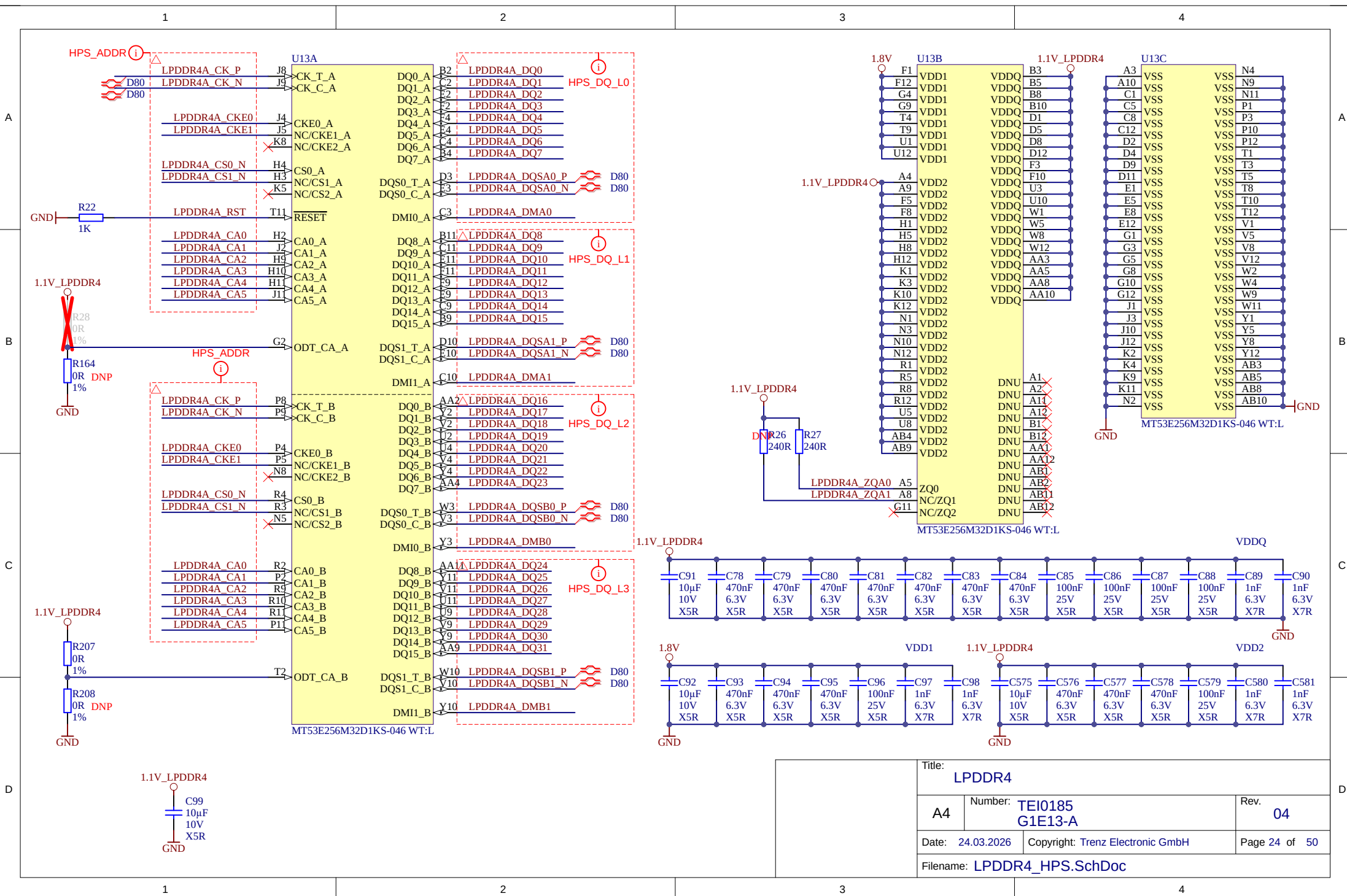
C

C

D

D





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A

A

B

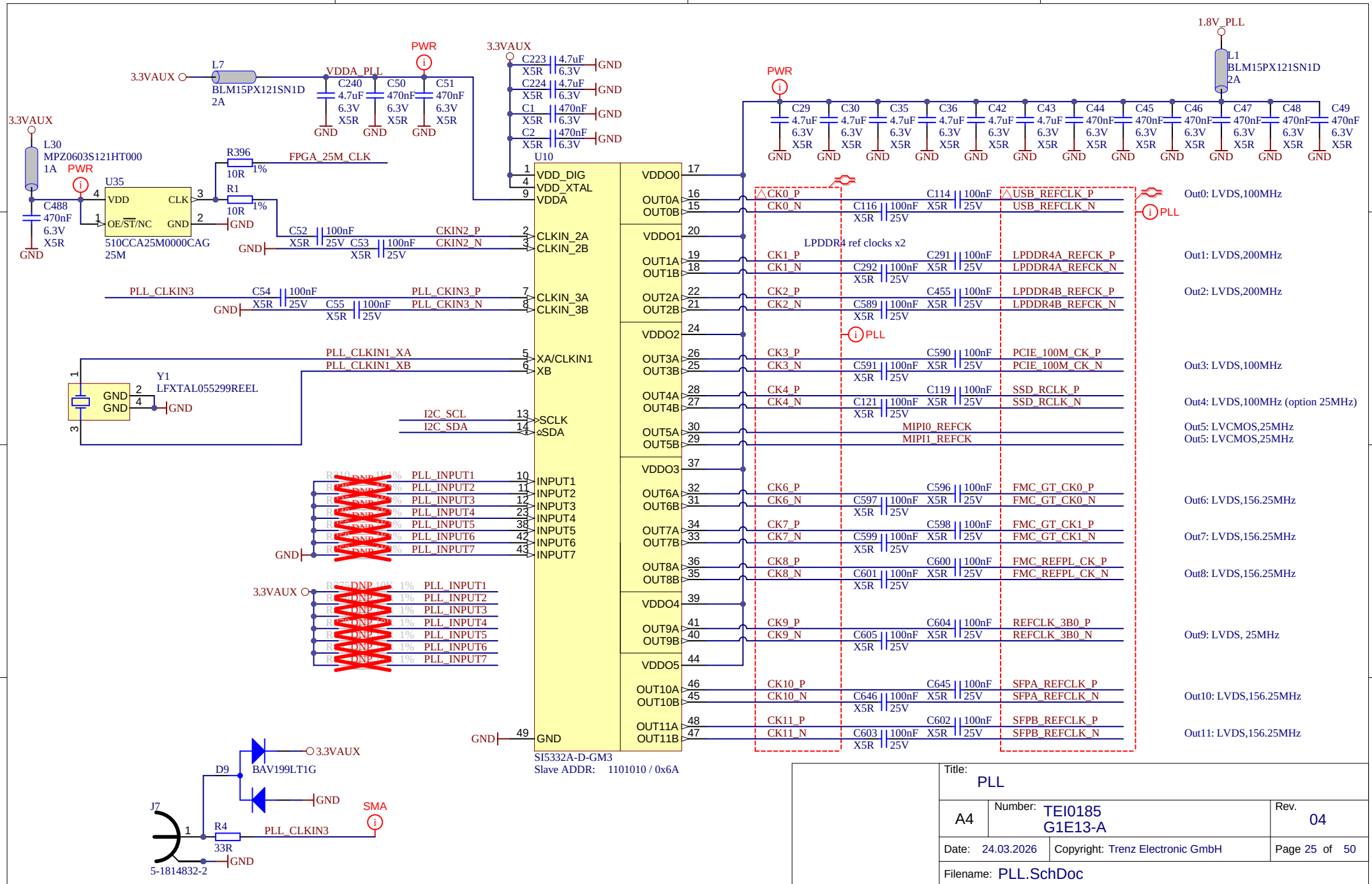
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C

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D

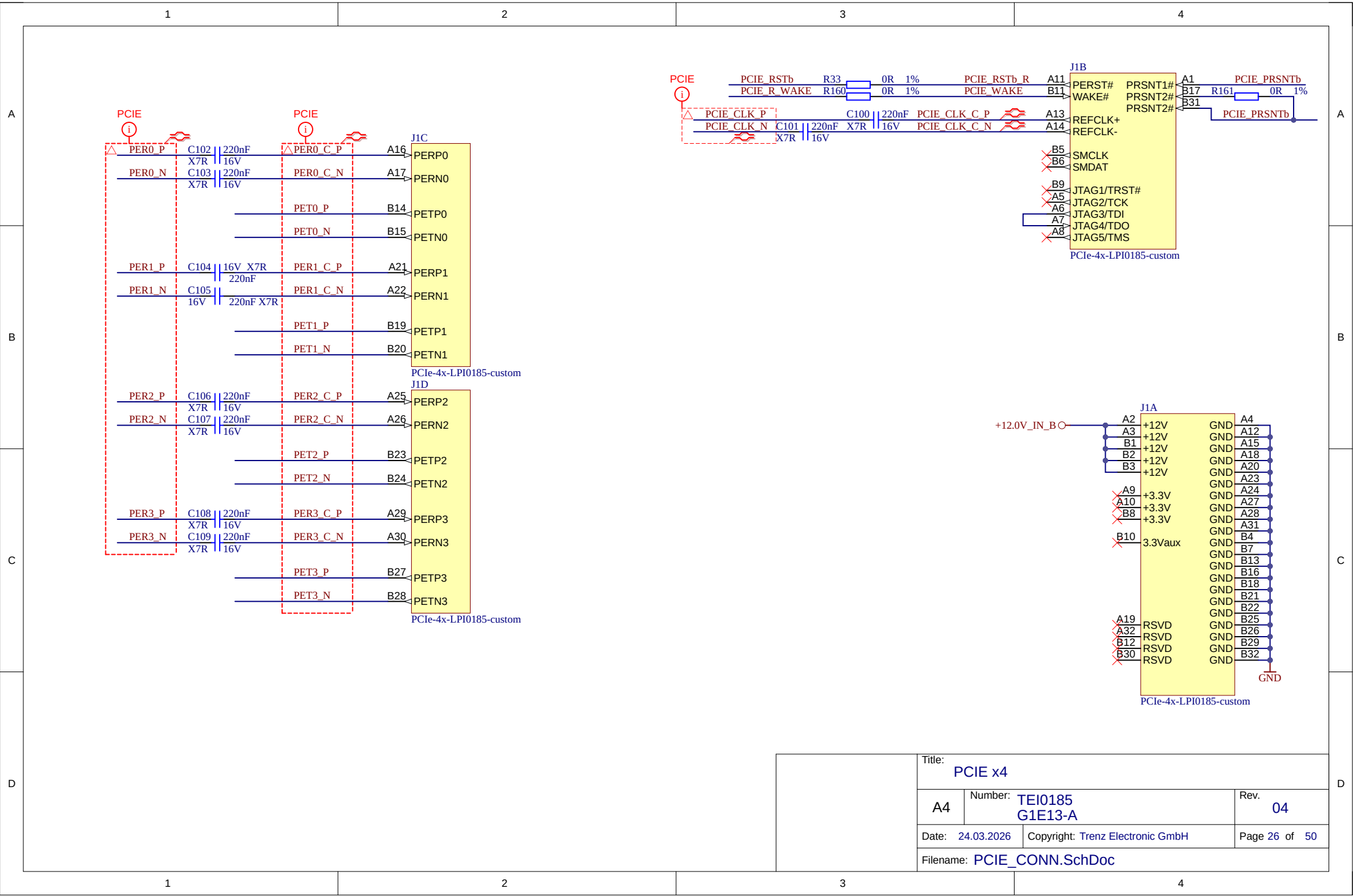


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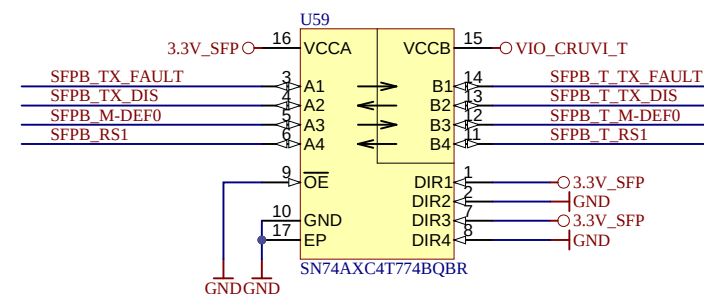
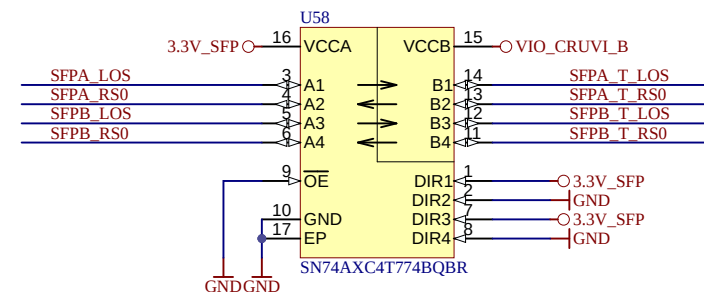
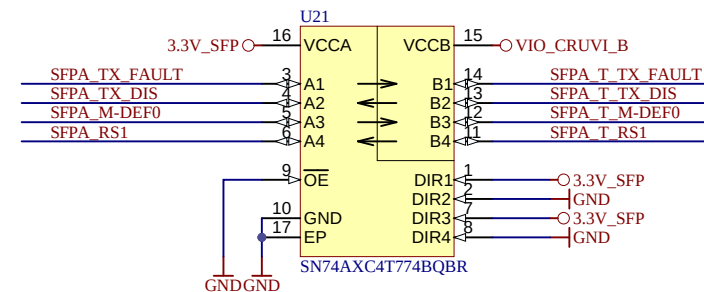
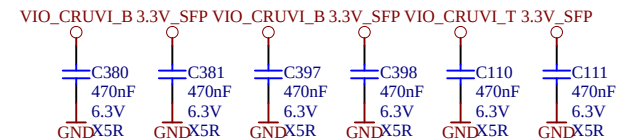
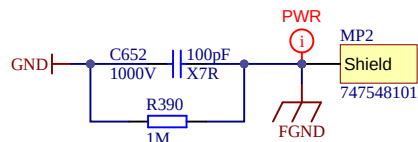
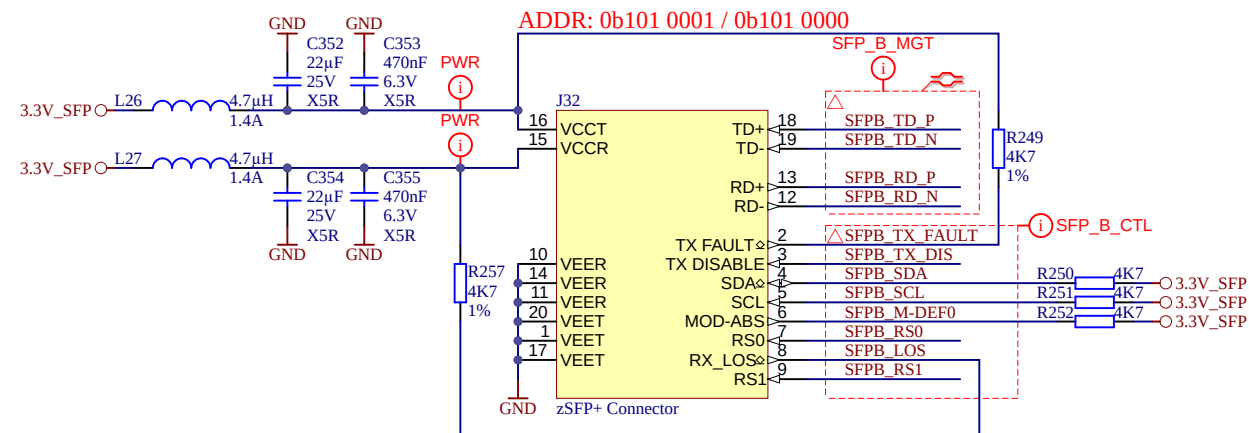
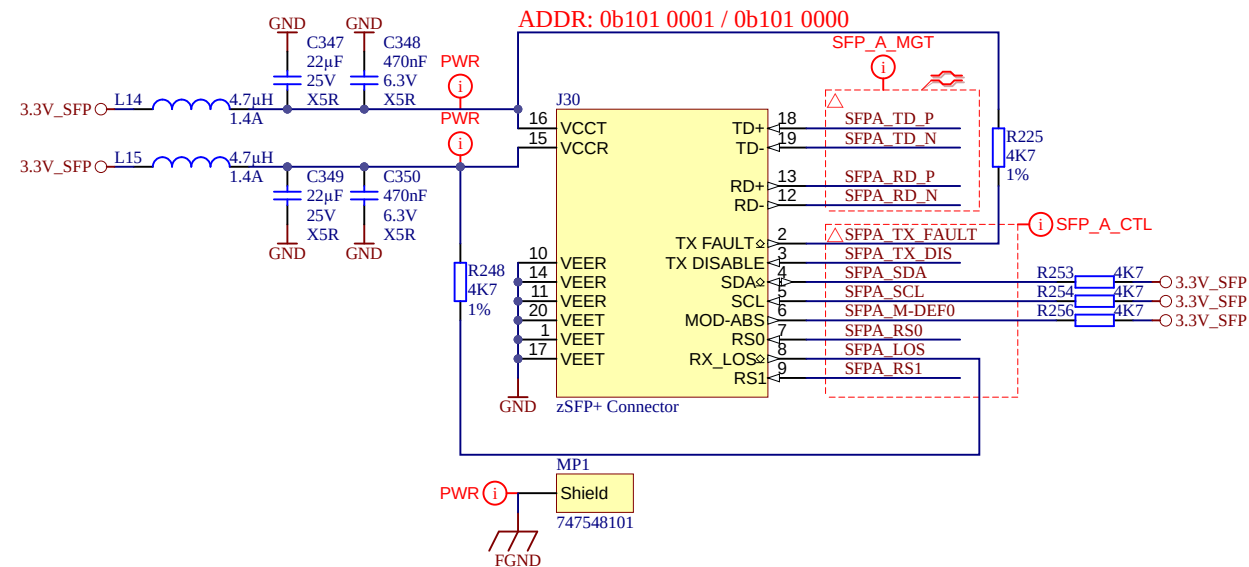
B

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C

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D



Title:

zSFP+

A4

Number: TEI0185
G1E13-A

Rev. 04

Date: 24.03.2026

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Filename: zSFP+.SchDoc

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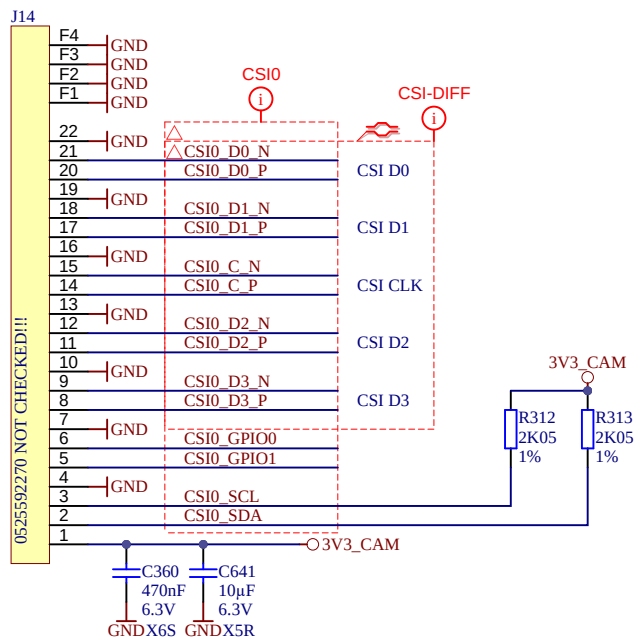
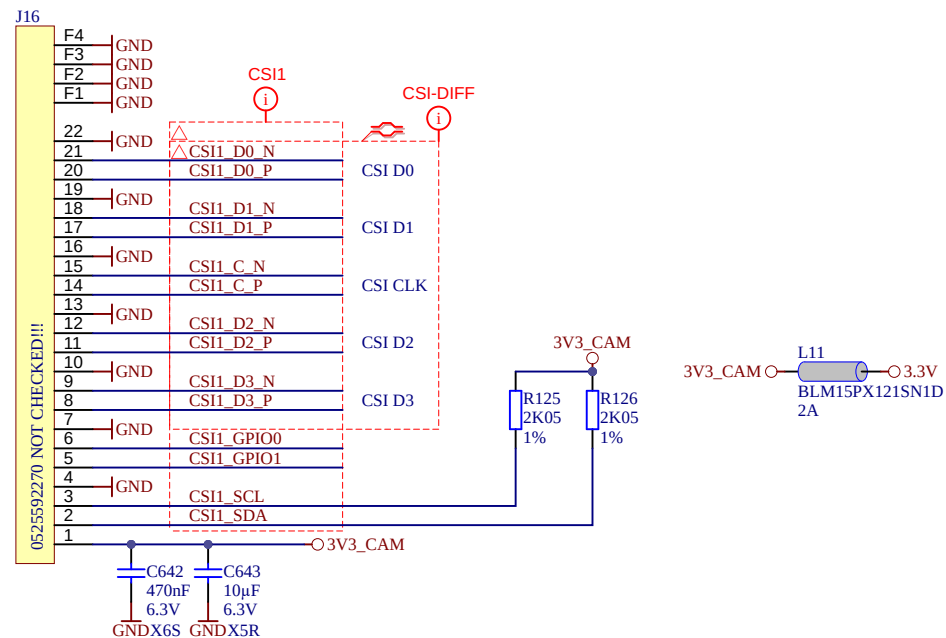
4

A

B

C

D

CSI-2 Camera
22pin, bottomCSI-2 Camera
22pin, bottom

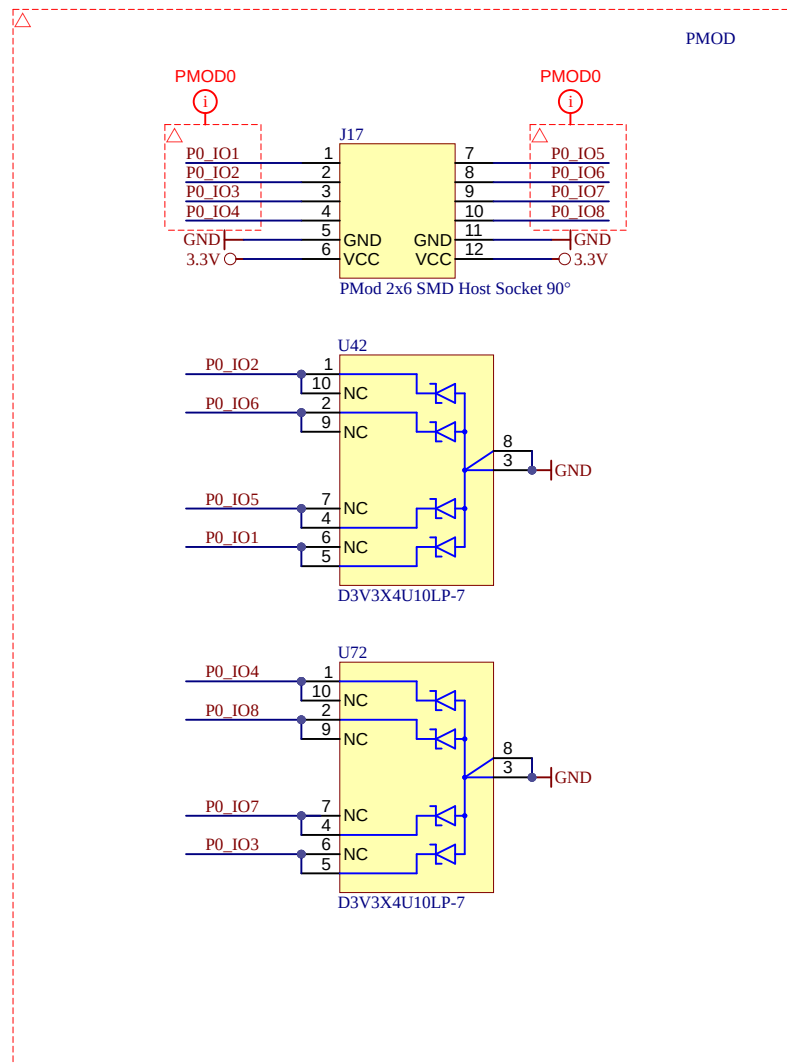
Title: MIPI CSI/DSI		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 28 of 50
Filename: CAM_0_1.SchDoc		

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			Title: Falcon - Pi HAT	
A4	Number: TEI0183 G1E13-A		Rev. 01	
Date:	2025-09-30	Copyright: Trenz Electronic GmbH		Page 29 of 34
Filename:		PMOD.SchDoc		

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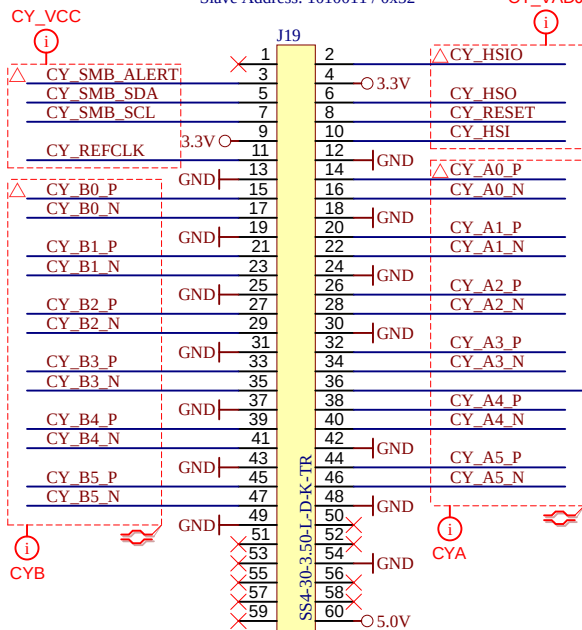
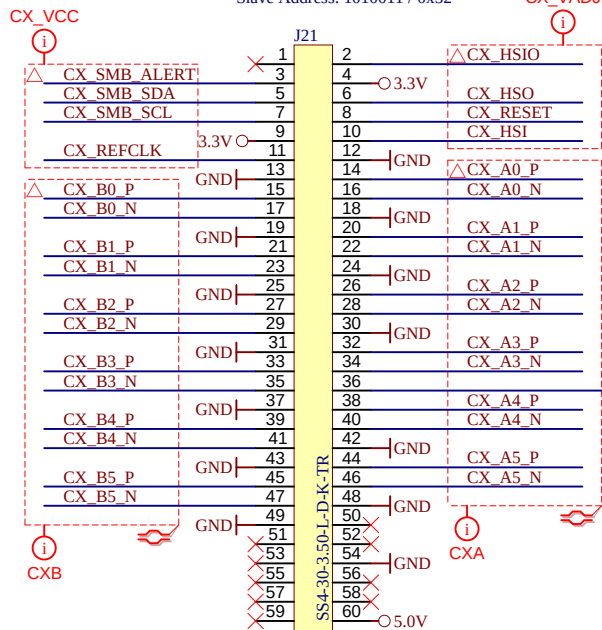
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Slave Address: 1010011 / 0x52

CX_VADJ

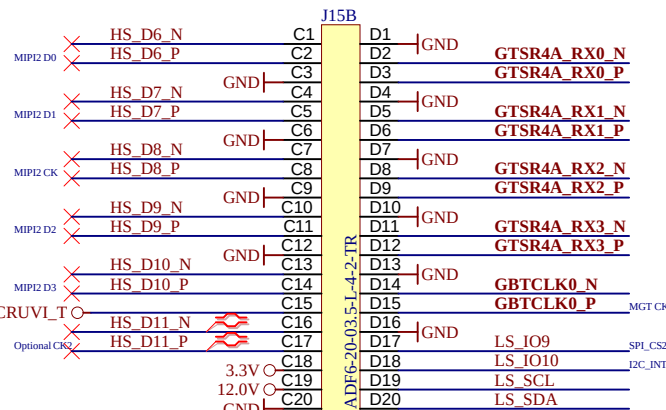
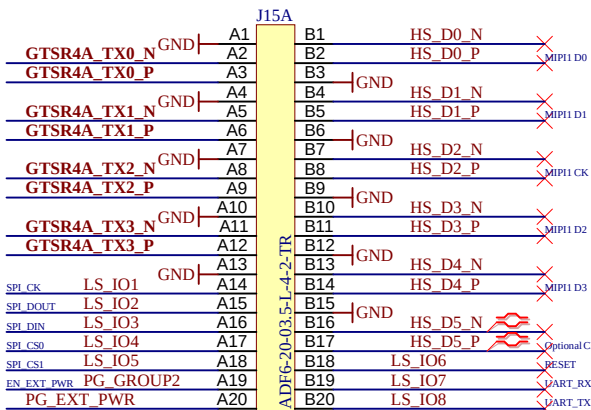
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CY_VADJ



VIO_CRUVI_T

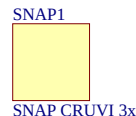
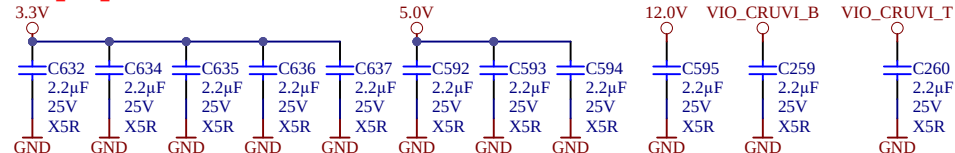
VIO_CRUVI_B



VIO_CRUVI_T

MGT CK

CHECK!!!! EN_EXT_PWR



SNAP CRUVI 3x

Title:

CRUVI

A4

Number: TEI0185
G1E13-A

Rev. 04

Date: 15.05.2026

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Filename: CRUVI.SchDoc

FMC_LA		J3B		FMC_LA	
LA00_P	G6	LA00_P_CC	LA19_P	LA19_P	
LA00_N	G7	LA00_N_CC	LA19_N	LA19_N	
LA01_P	D8	LA01_P_CC	LA20_P	LA20_P	
LA01_N	D9	LA01_N_CC	LA20_N	LA20_N	
LA02_P	H7	LA02_P	LA21_P	LA21_P	
LA02_N	H8	LA02_N	LA21_N	LA21_N	
LA03_P	G9	LA03_P	LA22_P	LA22_P	
LA03_N	G10	LA03_N	LA22_N	LA22_N	
LA04_P	H10	LA04_P	LA23_P	LA23_P	
LA04_N	H11	LA04_N	LA23_N	LA23_N	
LA05_P	D11	LA05_P	LA24_P	LA24_P	
LA05_N	D12	LA05_N	LA24_N	LA24_N	
LA06_P	C10	LA06_P	LA25_P	LA25_P	
LA06_N	C11	LA06_N	LA25_N	LA25_N	
LA07_P	H13	LA07_P	LA26_P	LA26_P	
LA07_N	H14	LA07_N	LA26_N	LA26_N	
LA08_P	G12	LA08_P	LA27_P	LA27_P	
LA08_N	G13	LA08_N	LA27_N	LA27_N	
LA09_P	D14	LA09_P	LA28_P	LA28_P	
LA09_N	D15	LA09_N	LA28_N	LA28_N	
LA10_P	C14	LA10_P	LA29_P	LA29_P	
LA10_N	C15	LA10_N	LA29_N	LA29_N	
LA11_P	H16	LA11_P	LA30_P	LA30_P	
LA11_N	H17	LA11_N	LA30_N	LA30_N	
LA12_P	G15	LA12_P	LA31_P	LA31_P	
LA12_N	G16	LA12_N	LA31_N	LA31_N	
LA13_P	D17	LA13_P	LA32_P	LA32_P	
LA13_N	D18	LA13_N	LA32_N	LA32_N	
LA14_P	C18	LA14_P	LA33_P	LA33_P	
LA14_N	C19	LA14_N	LA33_N	LA33_N	
LA15_P	H19	LA15_P			
LA15_N	H20	LA15_N			
LA16_P	G18	LA16_P			
LA16_N	G19	LA16_N			
LA17_P	D20	LA17_P_CC			
LA17_N	D21	LA17_N_CC			
LA18_P	C22	LA18_P_CC			
LA18_N	C23	LA18_N_CC			

ASP-184329-01, CC-HPC+-10

J3C

F4	HA00_P_CC	HA13_P	E12
F5	HA00_N_CC	HA13_N	E13
F6	HA01_P_CC	HA14_P	E15
F7	HA01_N_CC	HA14_N	E16
K8	HA02_P	HA15_P	E16
J6	HA02_N	HA15_N	E16
J7	HA03_P	HA16_P	E16
F7	HA03_N	HA16_N	E16
F8	HA04_P	HA17_P_CC	E17
E8	HA04_N	HA17_N_CC	E18
E9	HA05_P	HA18_P	E19
K10	HA05_N	HA18_N	E19
K11	HA06_P	HA19_P	E20
K11	HA06_N	HA19_N	E20
J9	HA07_P	HA20_P	E18
J10	HA07_N	HA20_N	E19
F10	HA08_P	HA21_P	E20
F11	HA08_N	HA21_N	E20
E9	HA09_P	HA22_P	E21
E10	HA09_N	HA22_N	E22
K13	HA10_P	HA23_P	E22
K14	HA10_N	HA23_N	E23
J14	HA11_P		
J13	HA11_N		
F14	HA12_P		
F14	HA12_N		

ASP-184329-01, CC-HPC+-10

J3D

K25	HB00_P_CC	HB13_P	E30
K26	HB00_N_CC	HB13_N	E31
J24	HB01_P	HB14_P	E34
J25	HB01_N	HB14_N	E35
F24	HB02_P	HB15_P	E33
F24	HB02_N	HB15_N	E34
E24	HB03_P	HB16_P	E34
E24	HB03_N	HB16_N	E35
F25	HB04_P	HB17_P_CC	E37
E24	HB04_N	HB17_N_CC	E36
E24	HB05_P	HB18_P	E37
E24	HB05_N	HB18_N	E37
K28	HB06_P_CC	HB19_P	E33
K29	HB06_N_CC	HB19_N	E34
J24	HB07_P	HB20_P	E37
J24	HB07_N	HB20_N	E38
F24	HB08_P	HB21_P	E36
F24	HB08_N	HB21_N	E37
E24	HB09_P		
E24	HB09_N		
K31	HB10_P		
K31	HB10_N		
J30	HB11_P		
J31	HB11_N		
F31	HB12_P		
F31	HB12_N		

ASP-184329-01, CC-HPC+-10

Title:

FMC+

A4

Number: TEI0185
G1E13-A

Rev.

04

Date: 24.03.2026

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Filename: FMC+_2.SchDoc

1

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A

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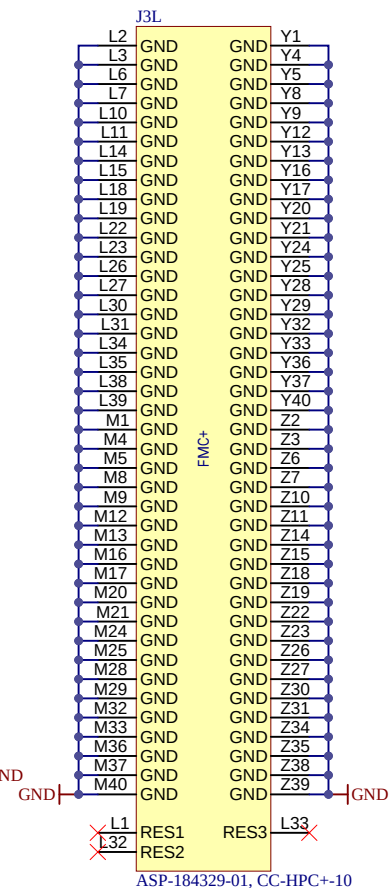
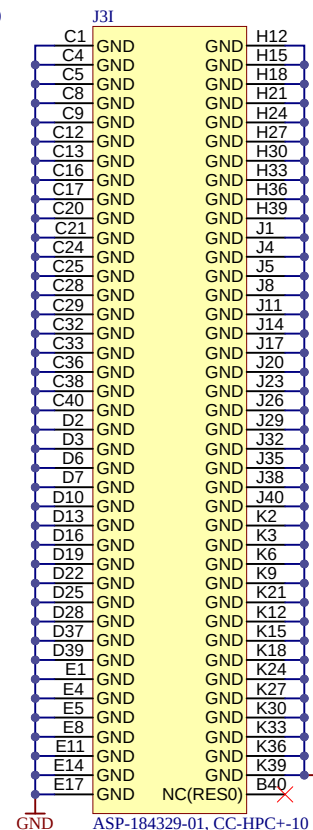
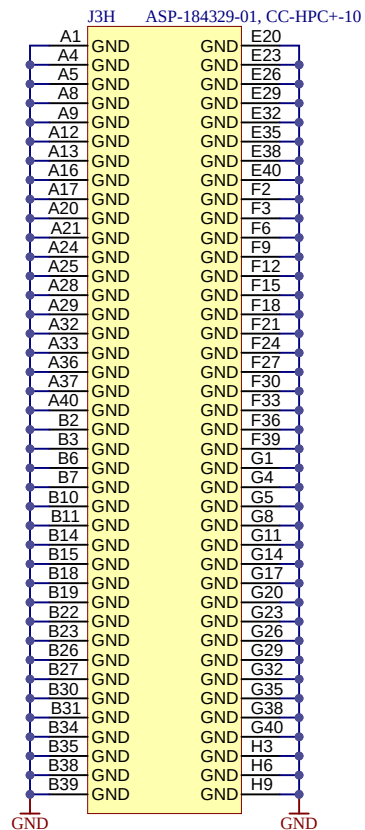
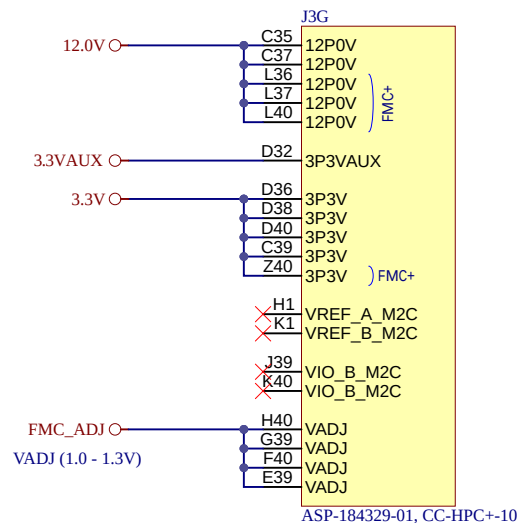
D

A

B

C

D



Title:

FMC+

A4

Number: TEI0185
G1E13-A

Rev. 04

Date: 24.03.2026

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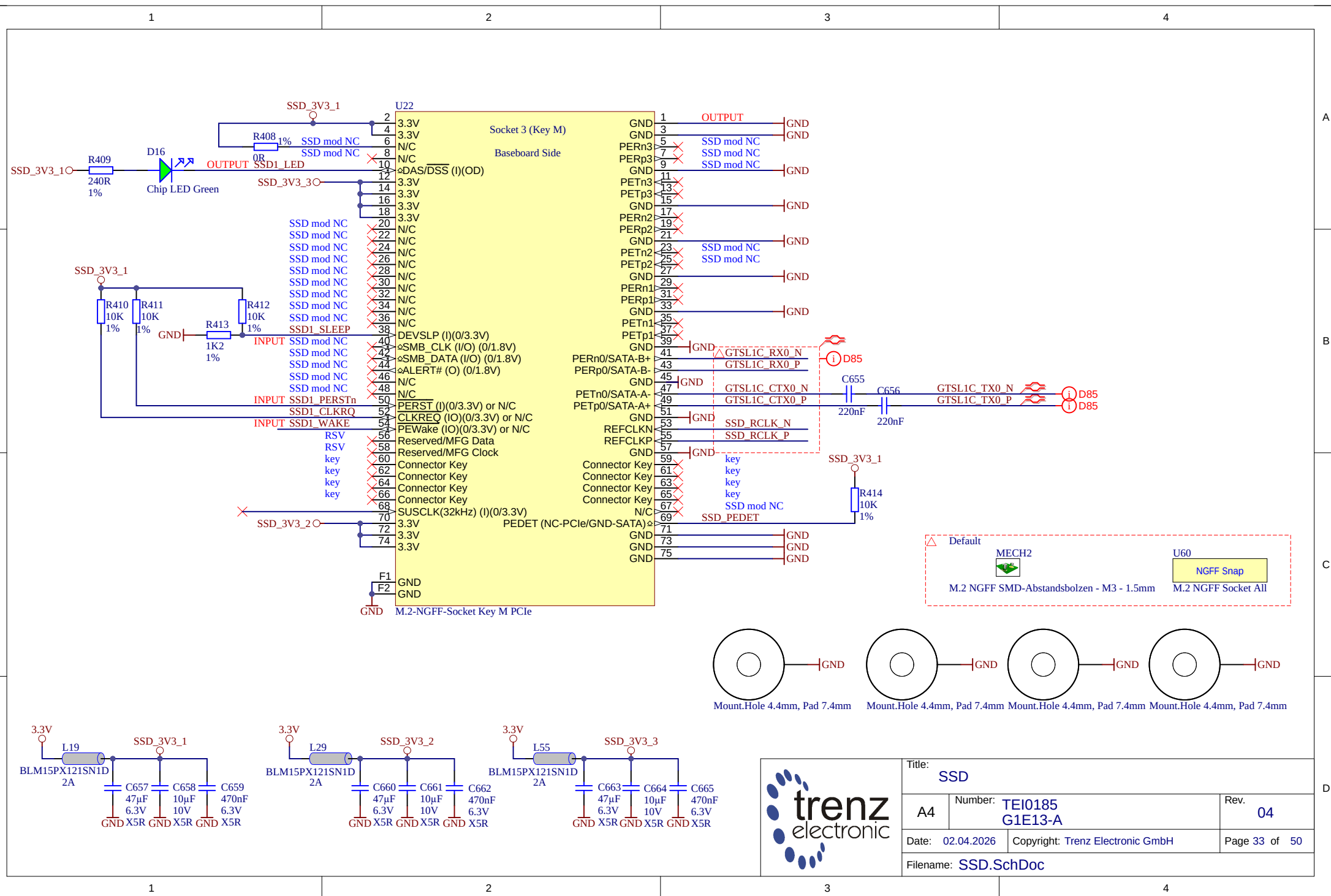
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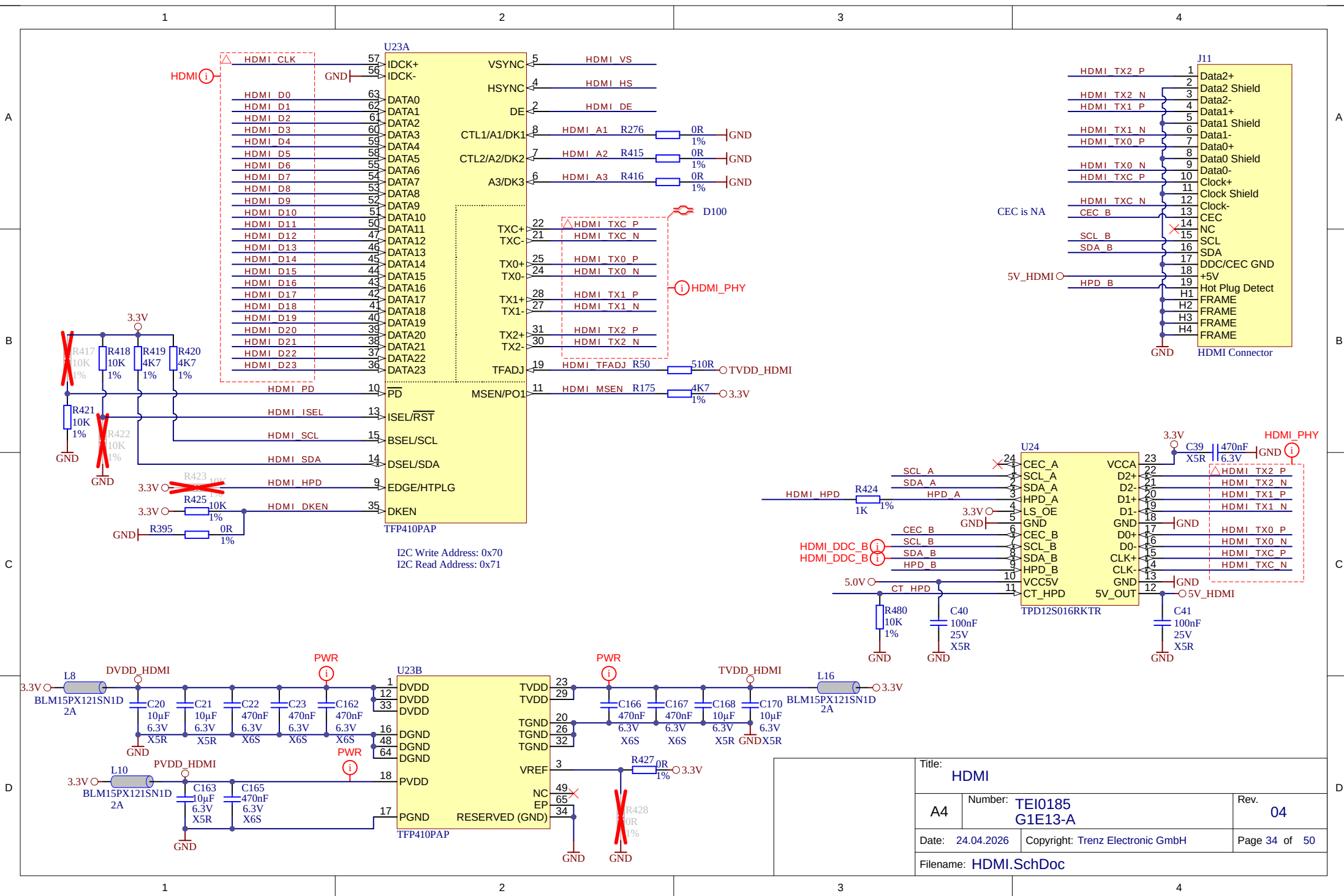
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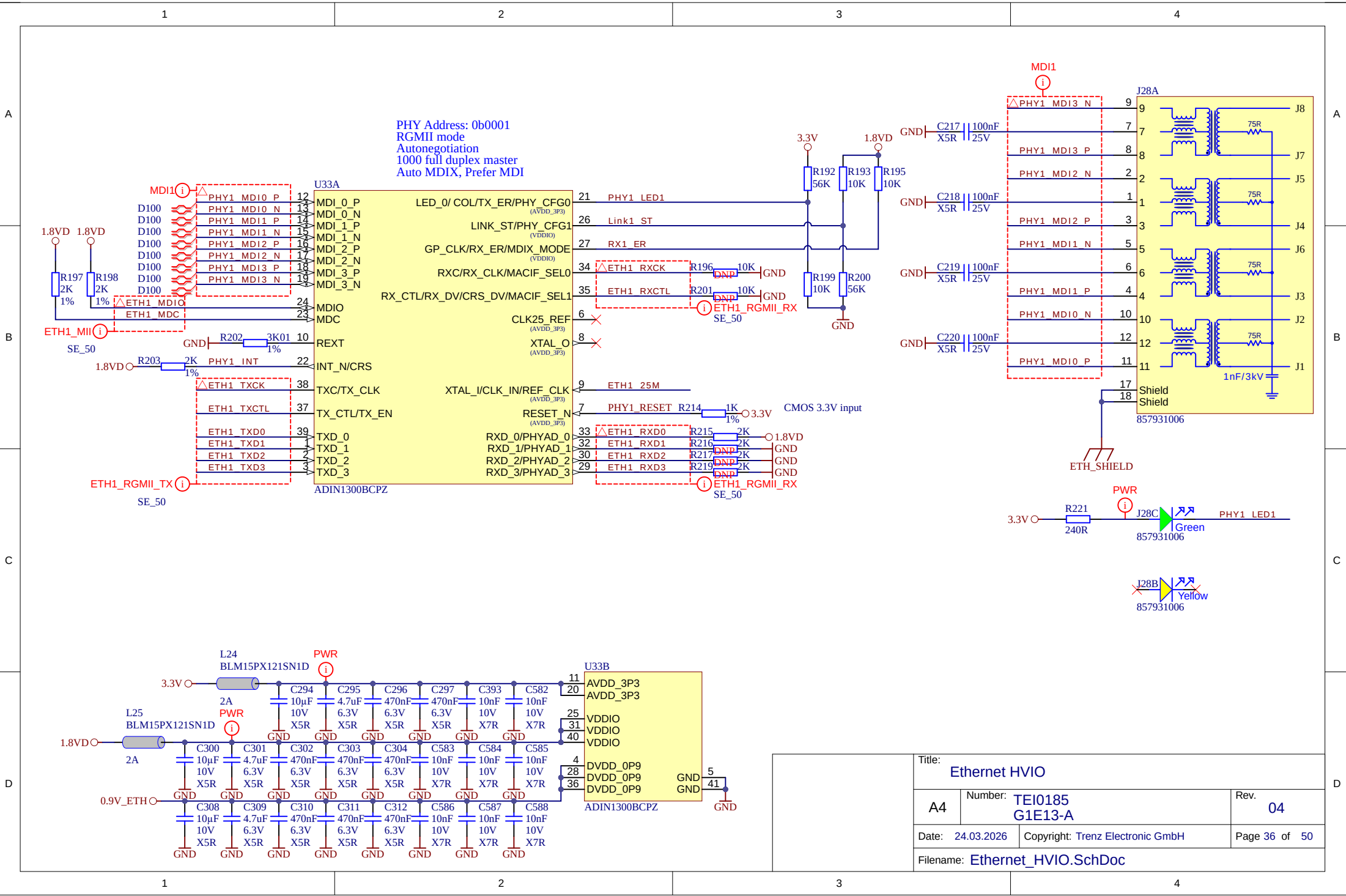
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Title: Ethernet HVIO		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 36 of 50
Filename: Ethernet_HVIO.SchDoc		

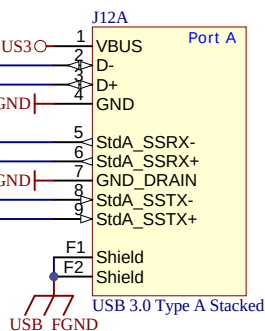
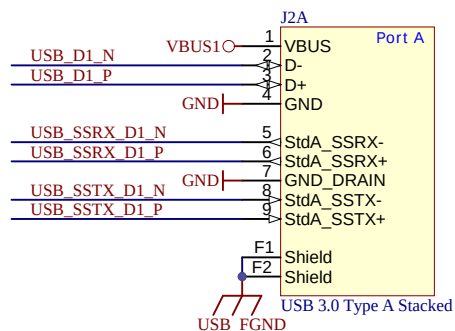
1

2

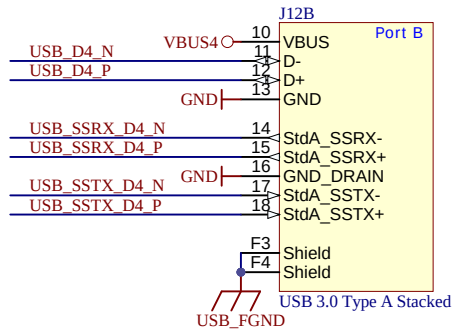
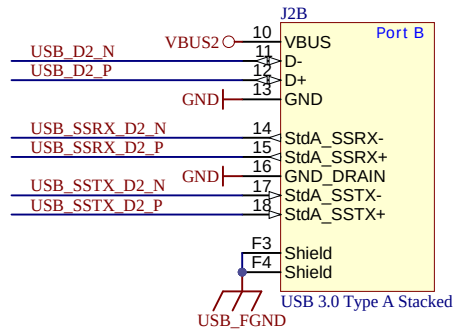
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4

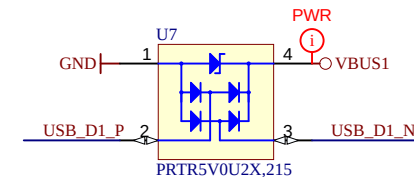
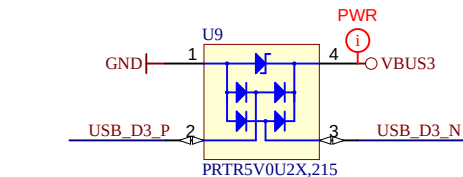
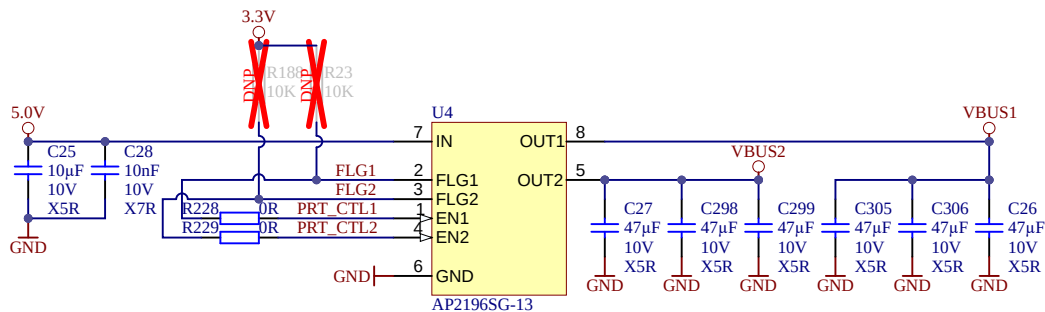
A



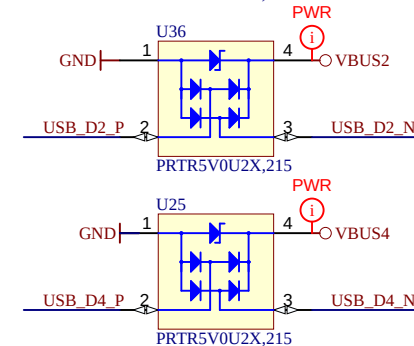
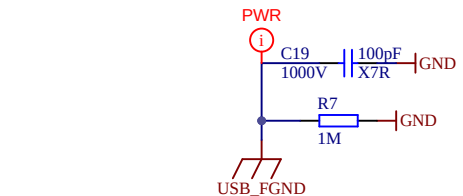
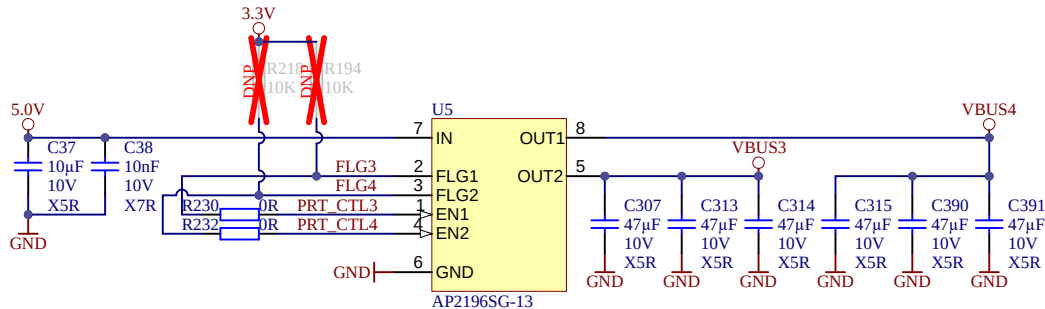
B



C



D



Title: USB		
A4	Number: TEI0185 G1E13-A	Rev. 04
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Filename: USB.SchDoc		

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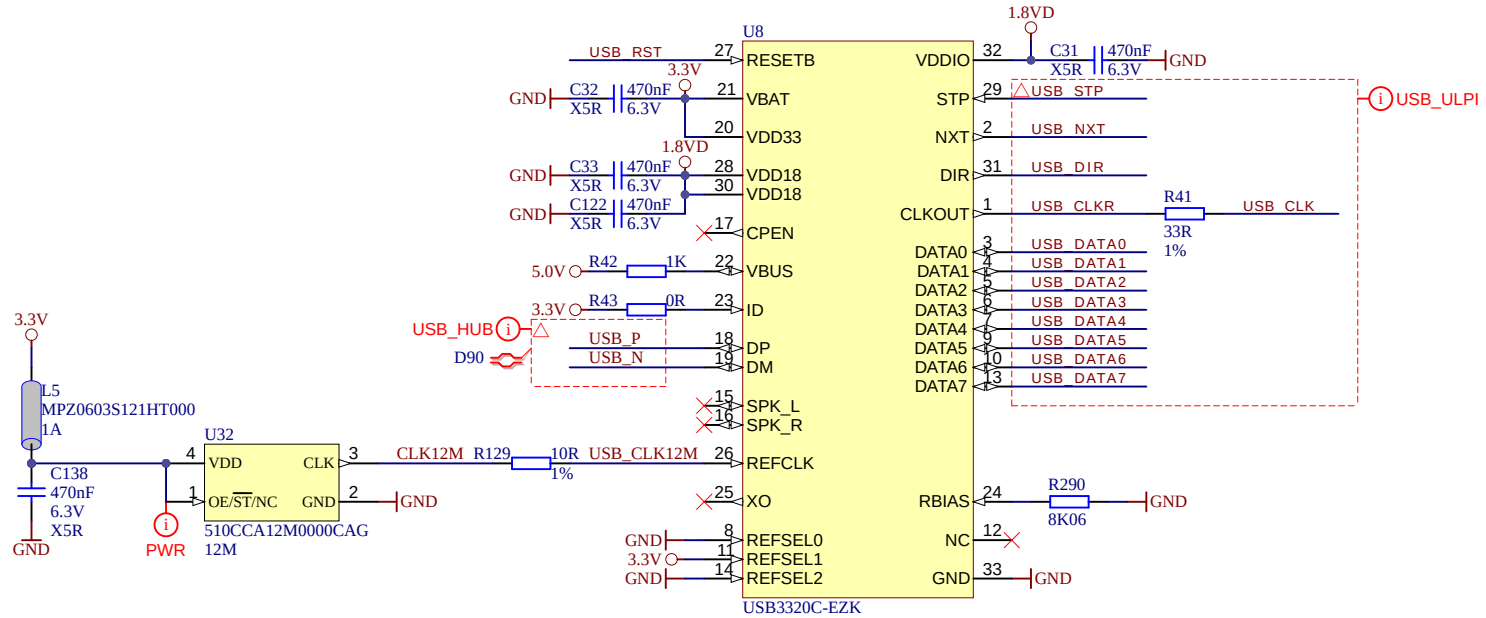
A

B

C

D

VBAT, VDD33, and VDD18 can be applied in any order.
The VDD18 supply must be turned on and stable before
the VDDIO supply is applied. This does not apply in
cases where the VDD18 and VDDIO are tied together.



Title: USB-PHY

A4

Number: TEI0185
G1E13-A

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Date: 27.04.2026

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Filename: USB-PHY.SchDoc

1

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A

A

B

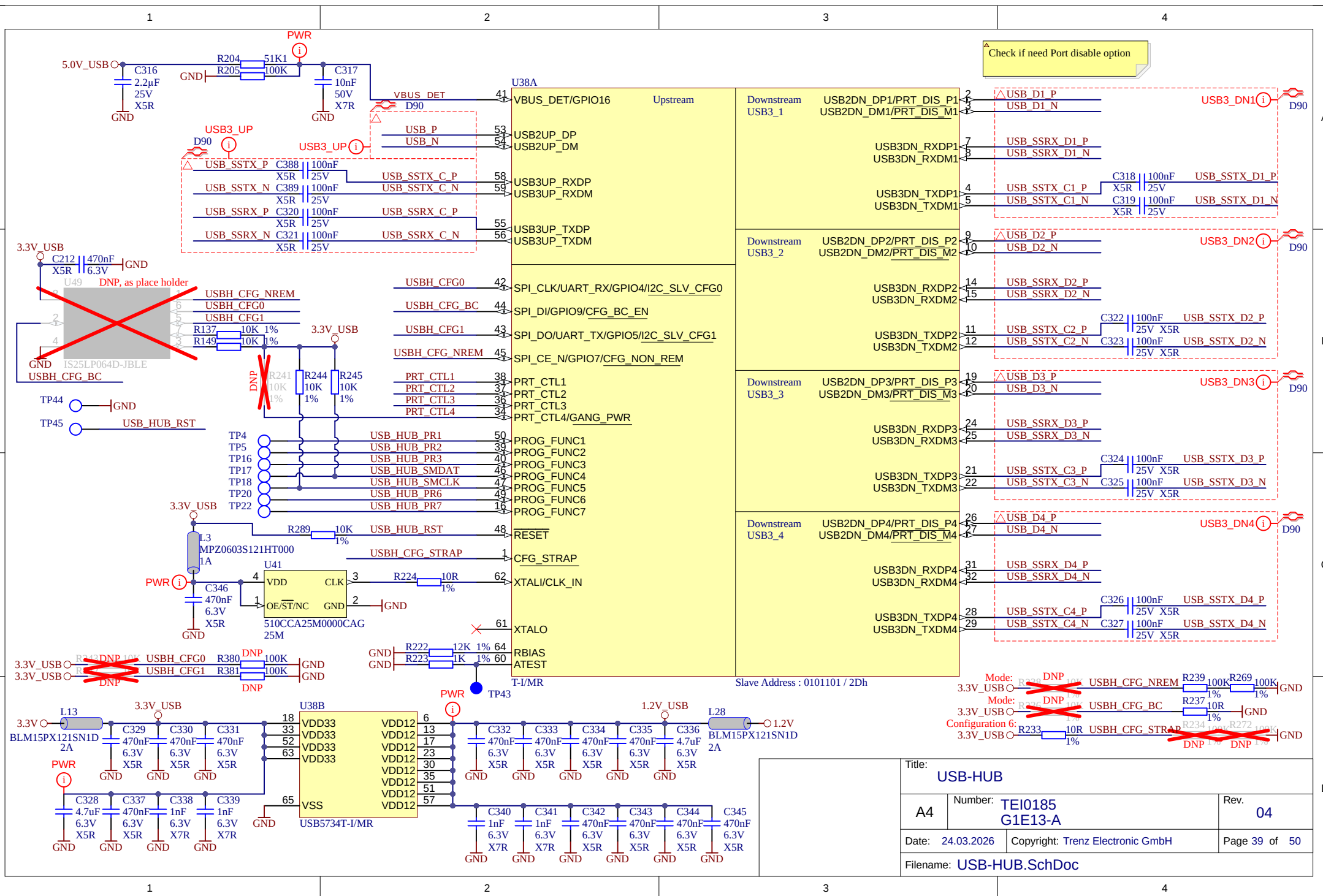
B

C

C

D

D



1

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A

A

B

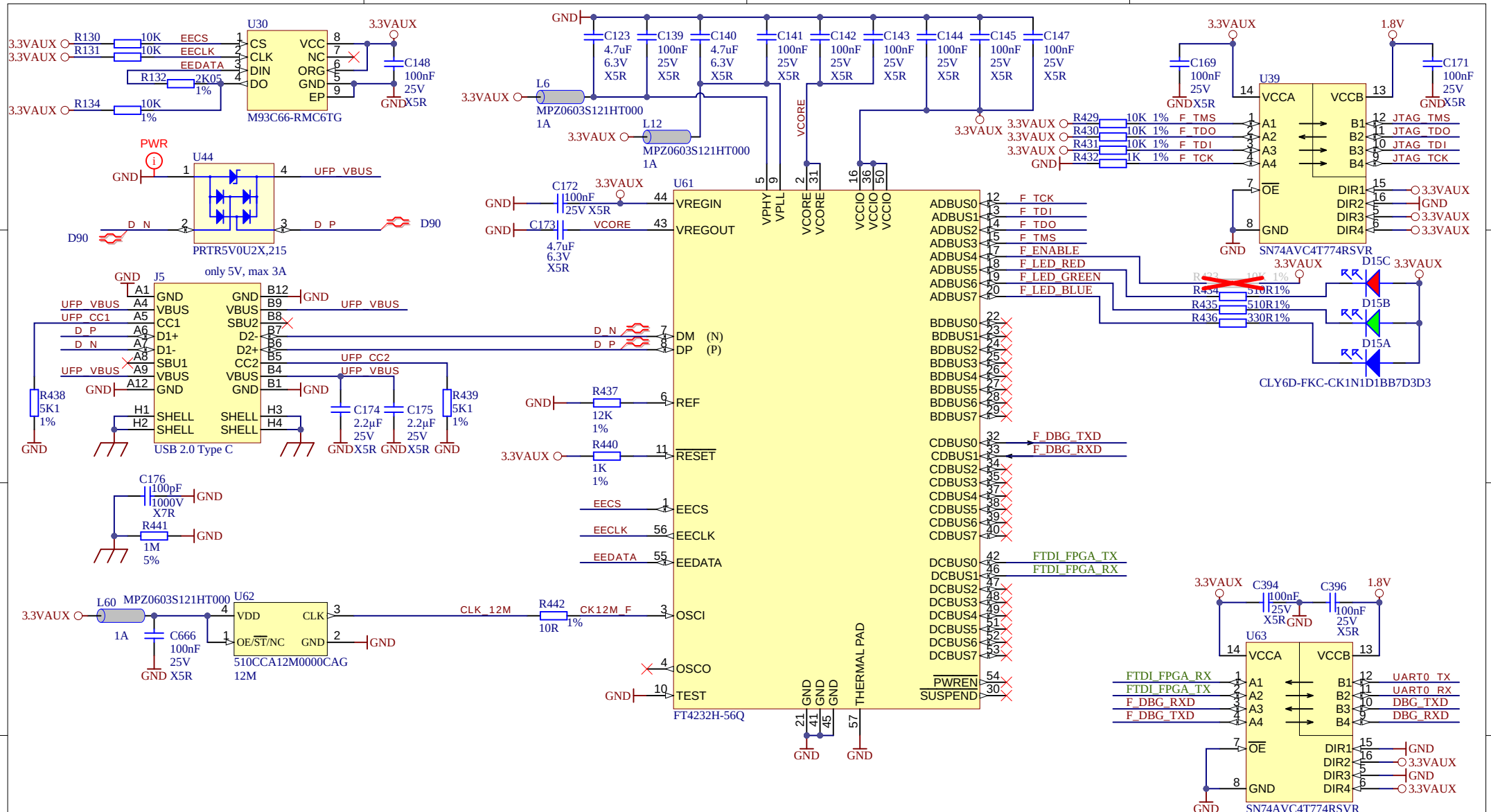
B

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C

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D



Title:

FTDI_JTAG

A4

Number:

TEI0185
G1E13-A

Rev.

04

Date: 19.05.2026

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Filename: FTDI_JTAG.SchDoc

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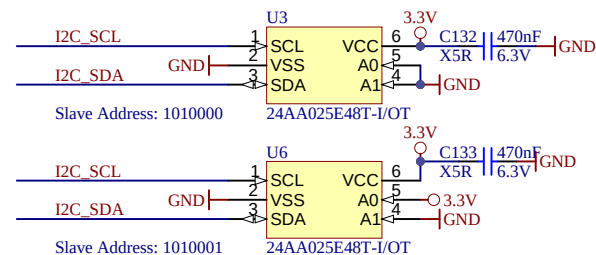
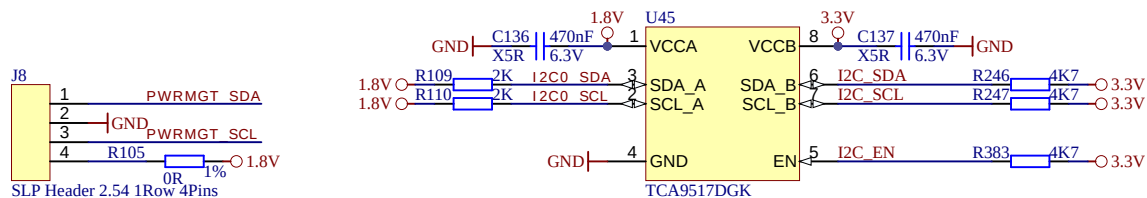
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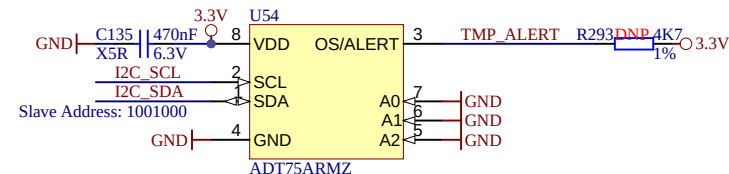
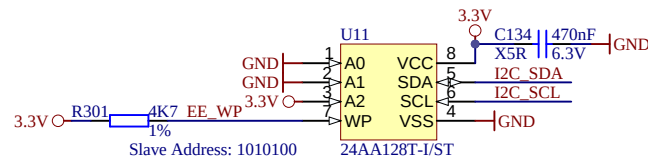
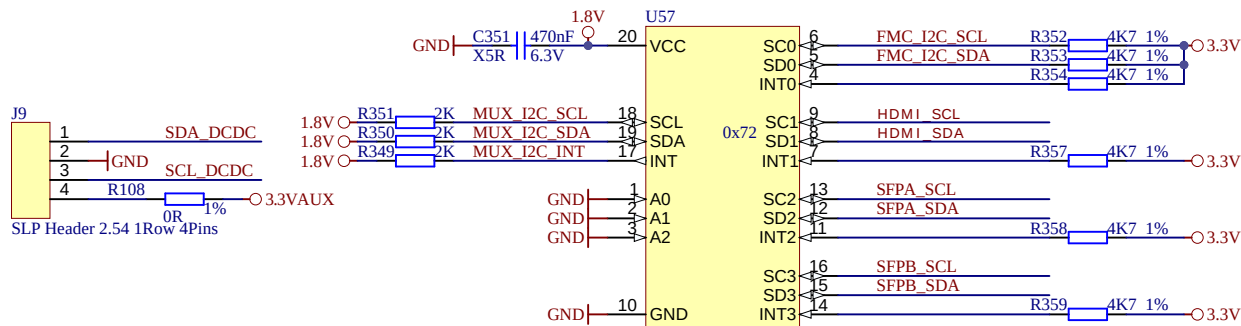
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A

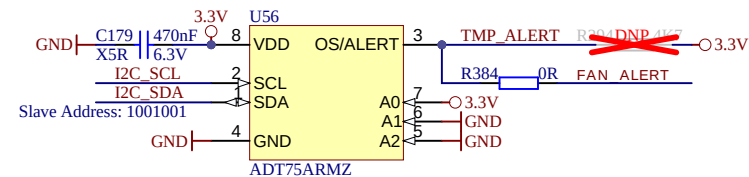
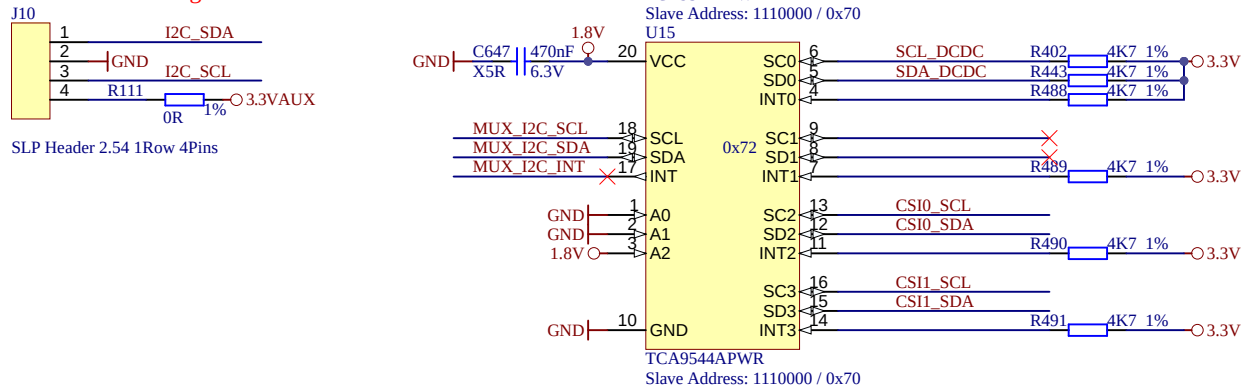


B

B

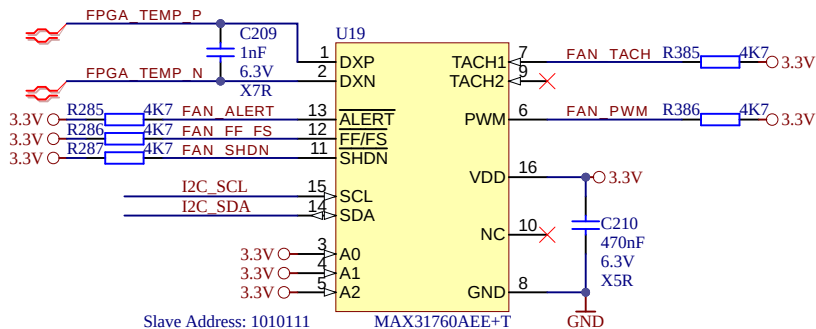


!!!! Check I2C voltage domain PLL



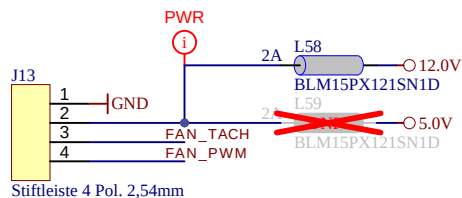
C

C



D

D



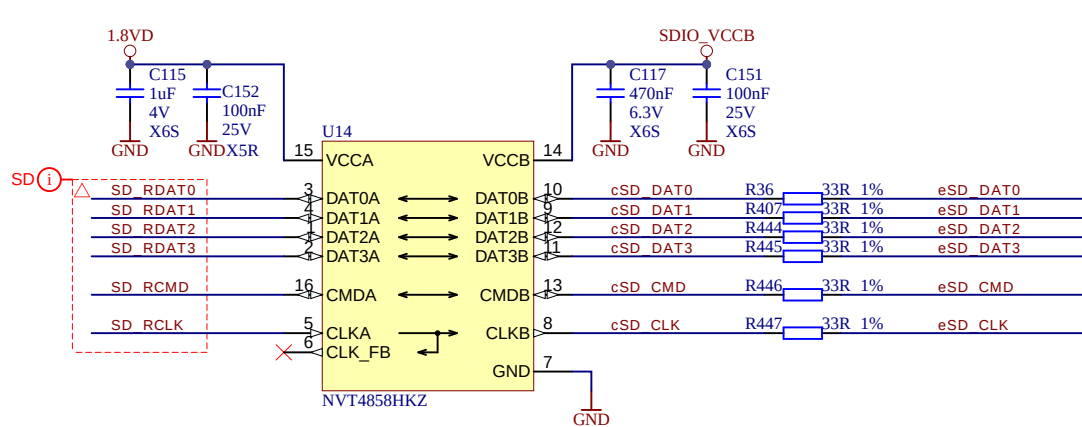
Title: Sensor_EEPROM		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 41 of 50
Filename: I2C_SENSOR_EEPROM.SchDoc		

1

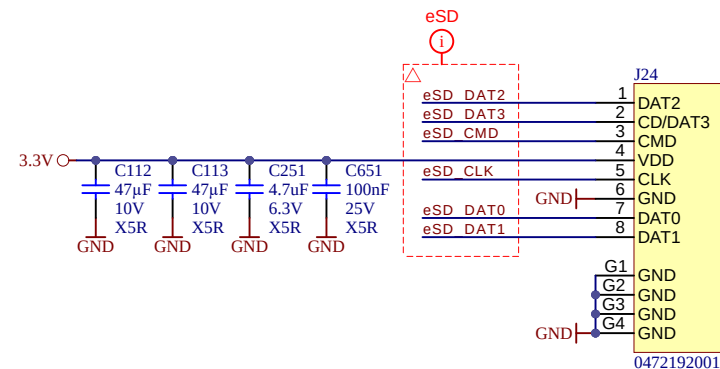
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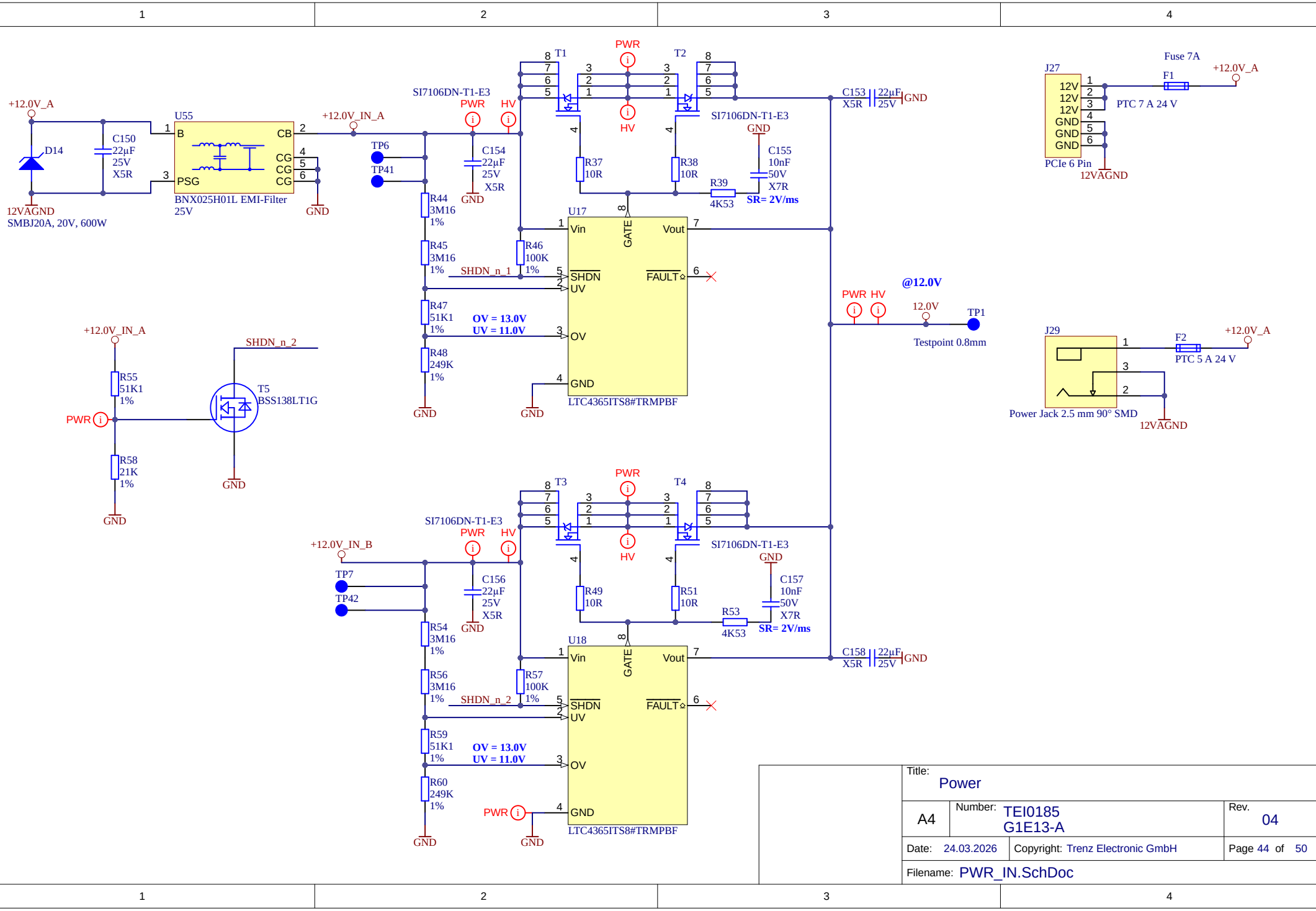
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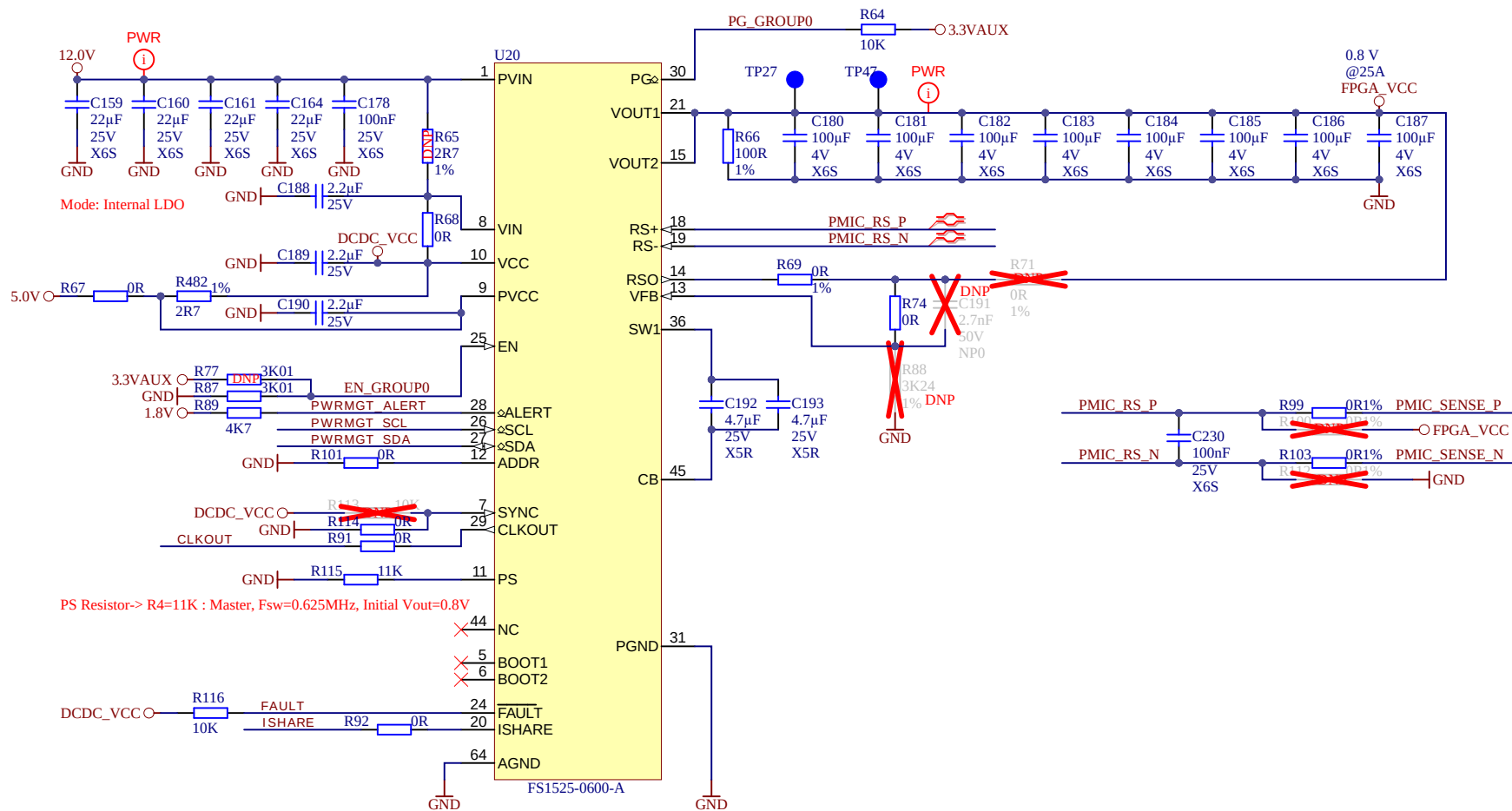
Support SD3.0



Title: SD		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 42 of 50
Filename: SD.schdoc		



Title: Power		
A4	Number: TEI0185 G1E13-A	Rev. 04
Date: 24.03.2026	Copyright: Trenz Electronic GmbH	Page 44 of 50
Filename: PWR_IN.SchDoc		



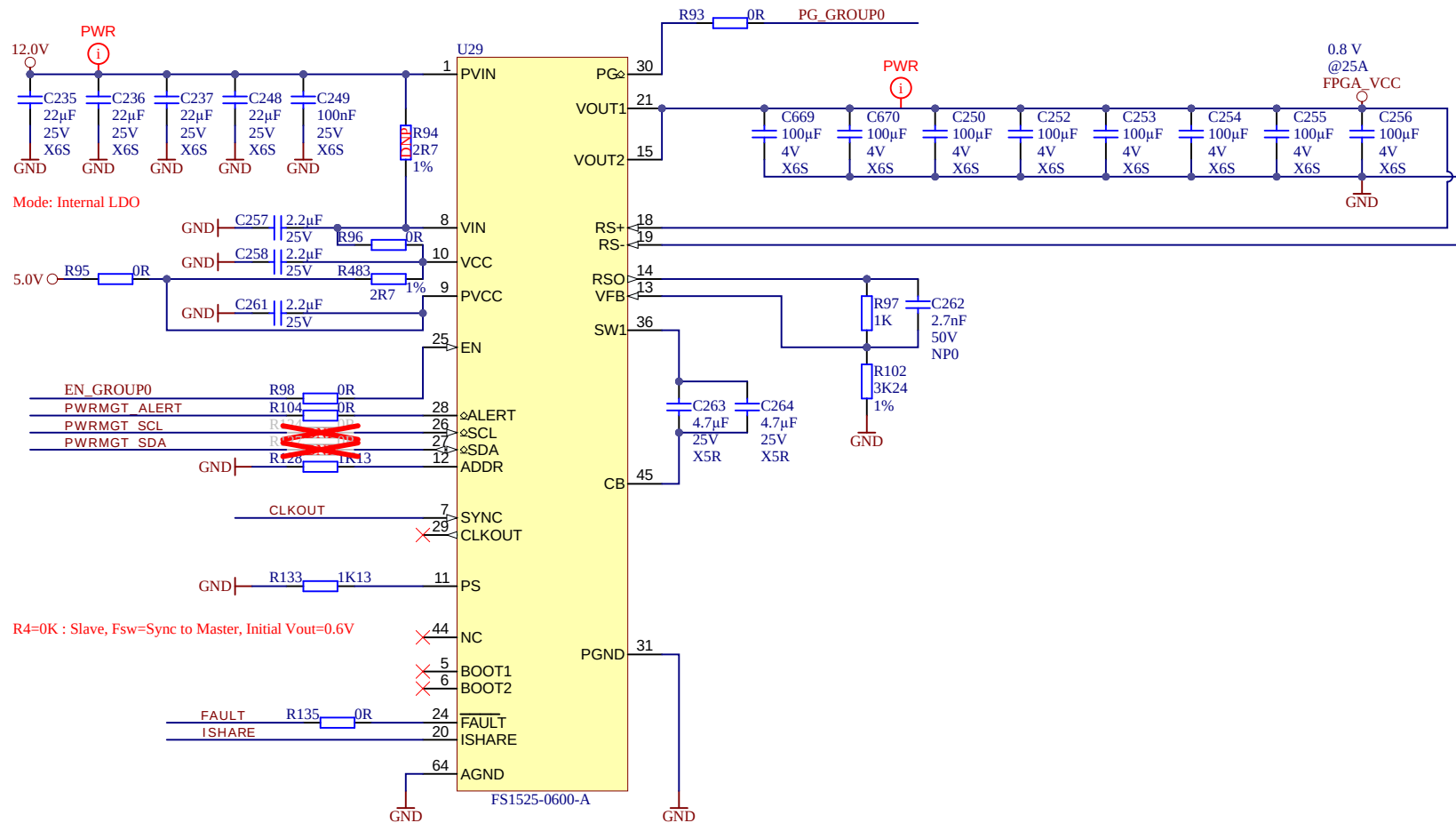
Title: Power		
A4	Number: TEI0185 G1E13-A	Rev. 04
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Filename: PWR_2.SchDoc		

1

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Title:

Power

A4

Number: TEI0185
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Rev.

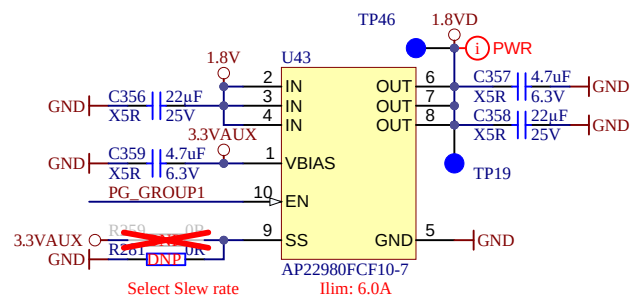
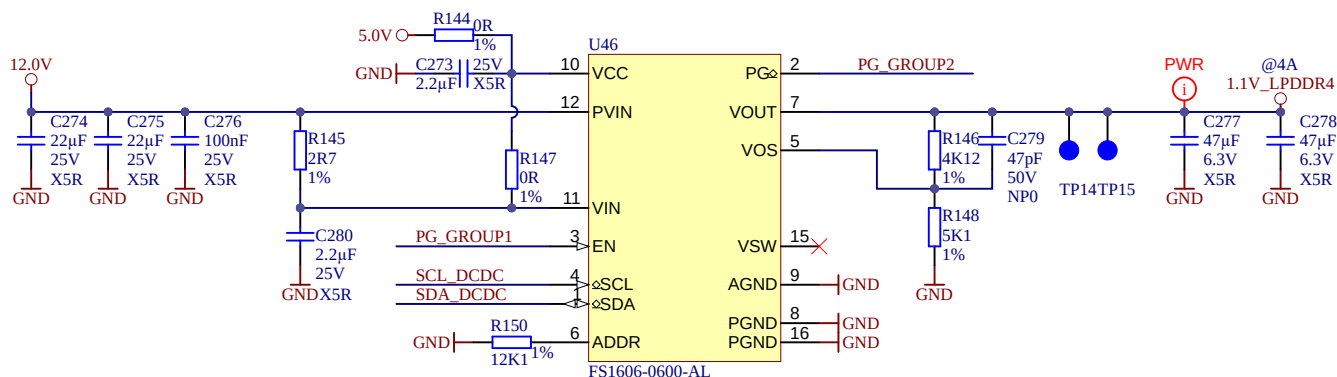
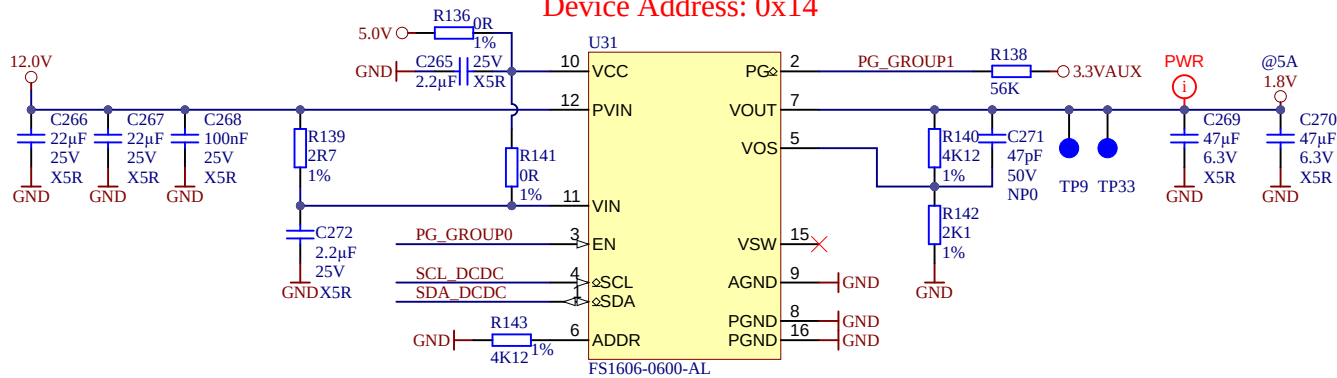
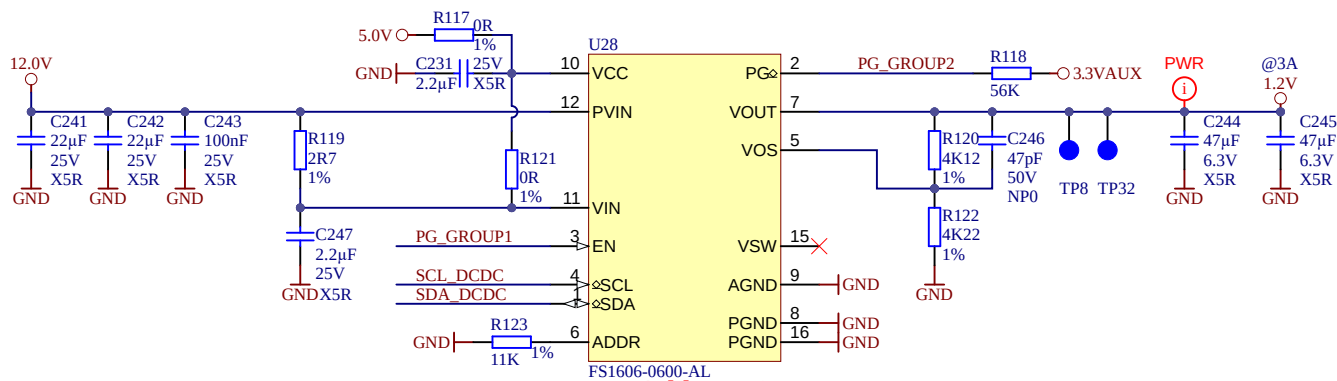
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Date: 08.04.2026

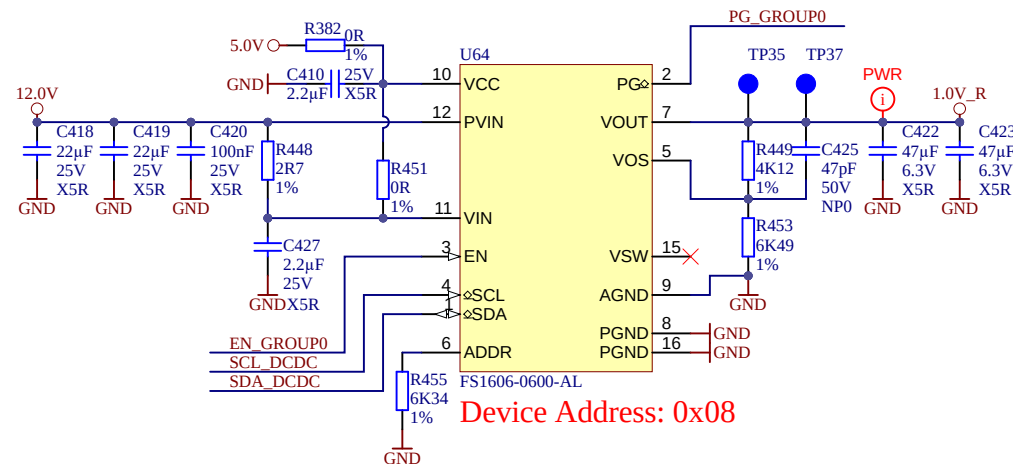
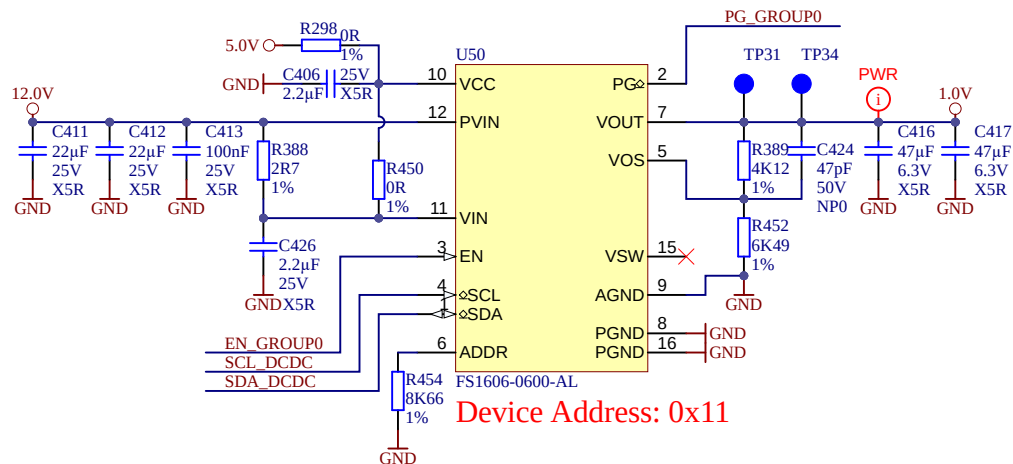
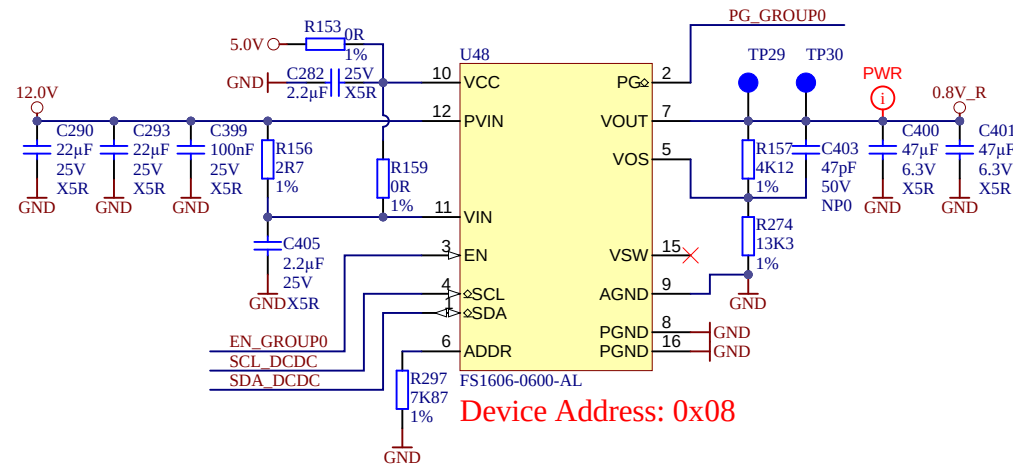
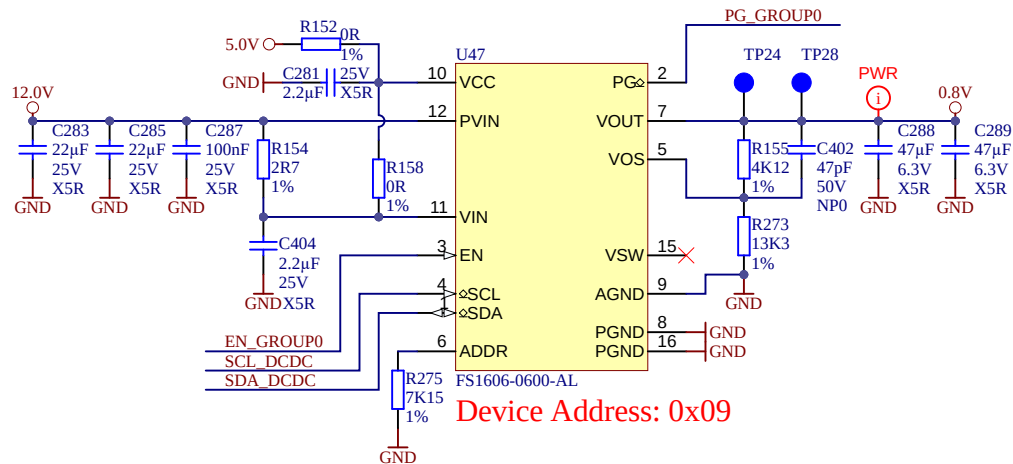
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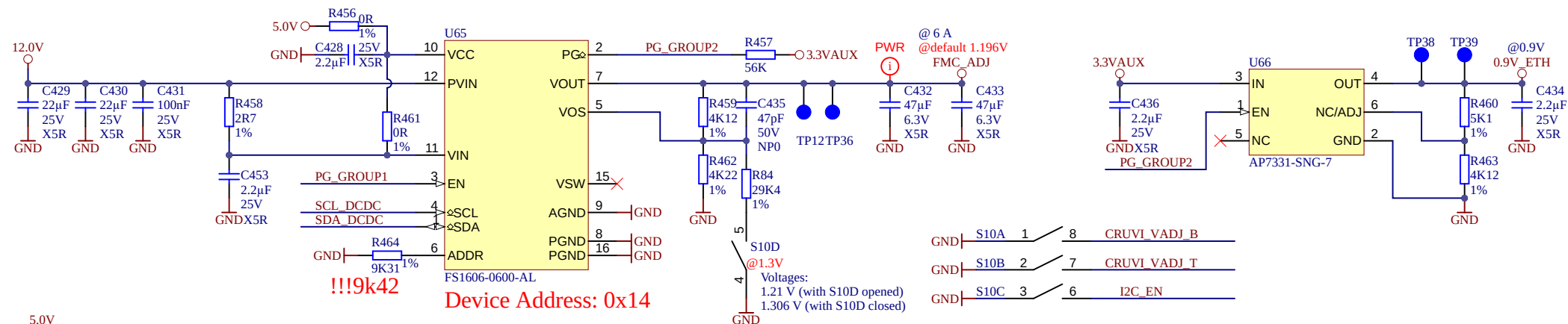
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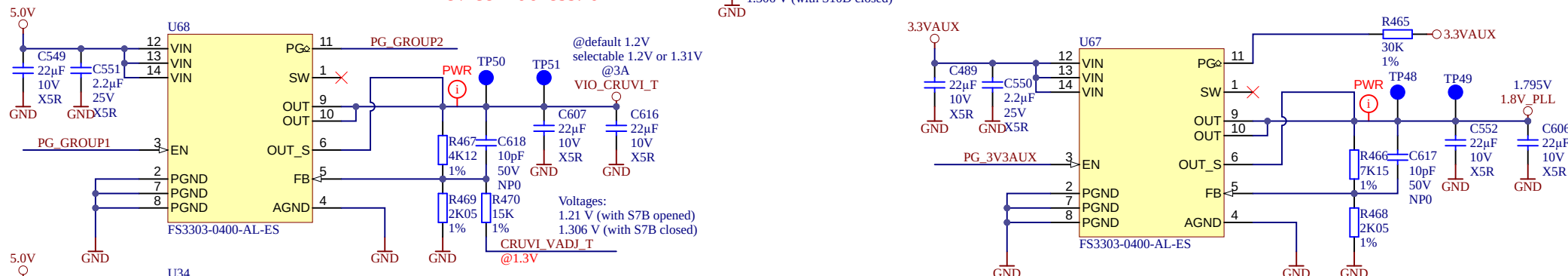
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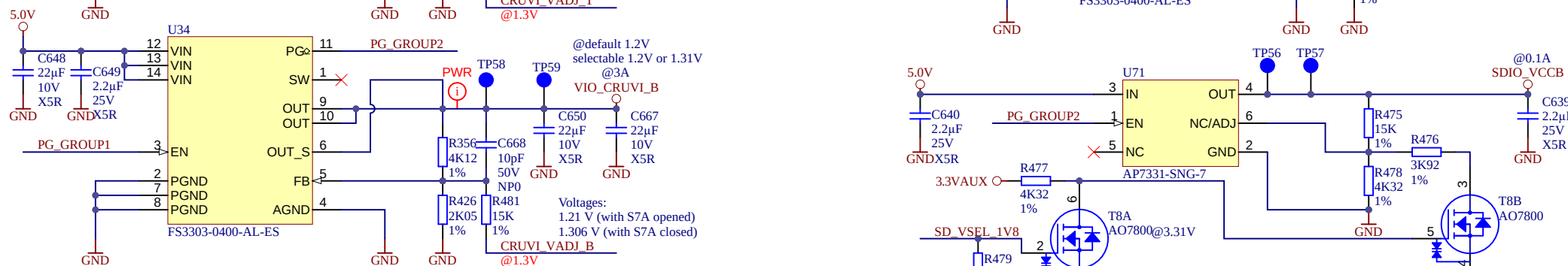
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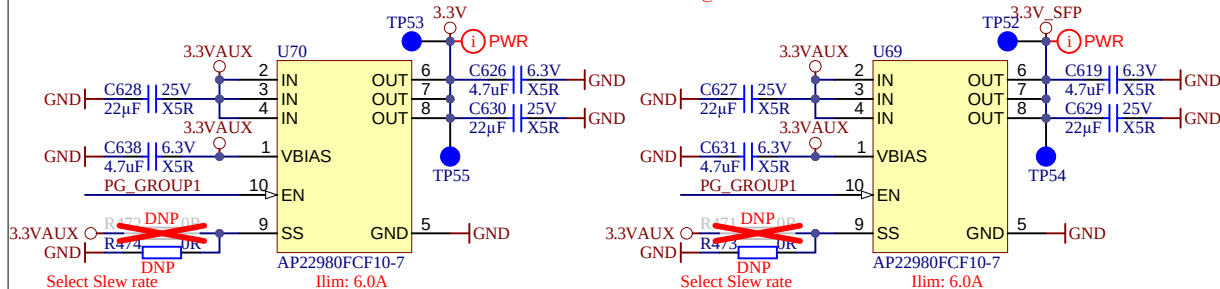
C

C



D

D



Power

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